# Job Skill Extraction Using NLP

Day-Wise Project Report

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Role: Data Analyst Intern

Project Duration Covered: Day 1 – Day 8

## Project Overview

This report documents the work completed across Days 1 through 7 of the Job Skill Extraction using NLP project, assigned as part of the internship task plan. The project's goal is to automatically identify and extract skills from unstructured job description text, moving from raw data collection through cleaning, taxonomy design, preprocessing, and a first rule-based extraction engine.

| | |
|---|---|
| **Project Title** | Job Skill Extraction Using NLP |
| **Days Covered** | Day 1 – Day 7 |
| **Core Stack** | Python, Pandas, NumPy, NLTK, spaCy, Scikit-learn, Jupyter Notebook, Google Colab |
| **Primary Data Field** | job_description |
| **Pipeline Stage Reached** | Rule-based dictionary skill extractor (baseline) |

## DAY 1 — Understanding the Problem

### Objective

To build a clear conceptual foundation for the project by understanding what a job description and a skill are, why automated skill extraction is valuable, and how the project's input/output pipeline is defined.

### Work Completed

- Studied the definitions of a job description and a skill in the context of NLP-based extraction.
- Reviewed why automatic skill extraction is useful (faster resume screening, skill-gap analysis, market trend tracking).
- Compared four NLP techniques relevant to the project: keyword extraction, entity extraction (NER), text classification, and information extraction, and identified how each applies here.
- Defined the project's input (raw job description text) and output (structured role + skill list) using a sample JSON schema.
- Prepared a formal Project Requirement Document covering Problem Statement, Business Objective, Technical Objective, Input, Expected Output, Technologies, and Success Criteria.

### Sample Input/Output Definition

```
Input:  "We require a Data Scientist with Python, Pandas, NumPy, Scikit-learn, SQL and AWS
experience."

Output:
{
  "role": "Data Scientist",
  "skills": ["Python", "Pandas", "NumPy", "Scikit-learn", "SQL", "AWS"]
}
```

### Tools & Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, NLTK, spaCy, Scikit-learn, Transformers, Matplotlib, Seaborn, Power BI/Tableau, Flask/FastAPI

### Outcome

A documented understanding of the problem space and a formal scope document that anchors every later stage of the project to a shared definition of success.

### Deliverable

**Project_Requirement_Document.pdf**

## DAY 2 — Dataset Collection

### Objective

To source a public dataset of job descriptions suitable for skill extraction and perform an initial inspection of its structure and quality.

### Work Completed

- Identified a suitable public dataset of job postings and downloaded it in CSV/JSON format.
- Loaded the dataset with pandas and inspected shape, column names, and sample rows.
- Checked for missing values across all fields, with particular attention to the job_description column.
- Identified duplicate job postings.
- Reviewed job_description text length distribution to understand typical posting size.
- Confirmed the recommended field set (job_id, job_title, company, location, job_description, experience, education, salary, job_type) and flagged job_description as the primary column for extraction.

### Initial Inspection Code

```
import pandas as pd
df = pd.read_csv("jobs.csv")
print(df.shape)
print(df.columns)
print(df.head())
print(df.isnull().sum())
```

### Tools & Technologies Used

Python, Pandas, Jupyter Notebook

### Outcome

A clean, inspected raw dataset with documented shape, missing-value counts, and duplicate counts, ready for exploratory analysis in Day 3.

### Deliverable and Screensorts

**raw_jobs.csv and dataset_analysis.ipynb**

1. MISSING VALUES

| | 0 |
|---|---|
| Company Profile | 15 |
| Experience | 0 |
| Job Id | 0 |
| Salary Range | 0 |
| location | 0 |
| Country | 0 |
| Qualifications | 0 |
| latitude | 0 |
| longitude | 0 |
| Company Size | 0 |
| Work Type | 0 |
| Preference | 0 |
| Contact Person | 0 |
| Contact | 0 |
| Job Posting Date | 0 |
| Job Title | 0 |
| Role | 0 |
| Job Description | 0 |
| Job Portal | 0 |
| Benefits | 0 |
| skills | 0 |
| Responsibilities | 0 |
| Company | 0 |

dtype: int64

| | Missing Count | Missing Percentage |
|---|---|---|
| Company Profile | 15 | 0.3 |
| Experience | 0 | 0.0 |
| Job Id | 0 | 0.0 |
| Salary Range | 0 | 0.0 |
| location | 0 | 0.0 |
| Country | 0 | 0.0 |
| Qualifications | 0 | 0.0 |
| latitude | 0 | 0.0 |
| longitude | 0 | 0.0 |
| Company Size | 0 | 0.0 |
| Work Type | 0 | 0.0 |
| Preference | 0 | 0.0 |
| Contact Person | 0 | 0.0 |
| Contact | 0 | 0.0 |
| Job Posting Date | 0 | 0.0 |
| Job Title | 0 | 0.0 |
| Role | 0 | 0.0 |
| Job Description | 0 | 0.0 |
| Job Portal | 0 | 0.0 |
| Benefits | 0 | 0.0 |
| skills | 0 | 0.0 |
| Responsibilities | 0 | 0.0 |
| Company | 0 | 0.0 |

## DAY 3 — Exploratory Data Analysis

### Objective

To analyze the collected job dataset statistically and visually before cleaning, to understand its composition and surface data-quality issues.

### Work Completed

- Calculated total number of jobs and number of unique job titles.
- Identified the most common job roles and job counts by location.
- Computed average, minimum, and maximum job_description length.
- Quantified missing values and duplicate records across the dataset.
- Built visualizations: top job titles, jobs by location, description-length distribution, missing-value summary, and top companies.

### Description Length Analysis

```
df["description_length"] = df["job_description"].str.len()
df["description_length"].describe()
```

## Tools & Technologies Used

Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

## Outcome

A statistical and visual profile of the dataset that informed the cleaning rules applied in Day 4 (e.g. handling outlier-length descriptions, duplicates, and missing text).

## Deliverable

### EDA_Report.ipynb

| | 0 |
|---|---|
| job_id | float64 |
| experience | object |
| qualifications | object |
| salary_range | object |
| location | object |
| country | object |
| latitude | float64 |
| longitude | float64 |
| work_type | object |
| company_size | int64 |
| job_posting_date | object |
| preference | object |
| contact_person | object |
| contact | object |
| job_title | object |
| role | object |
| job_portal | object |
| job_description | object |
| benefits | object |
| skills | object |
| responsibilities | object |
| company | object |
| company_profile | object |

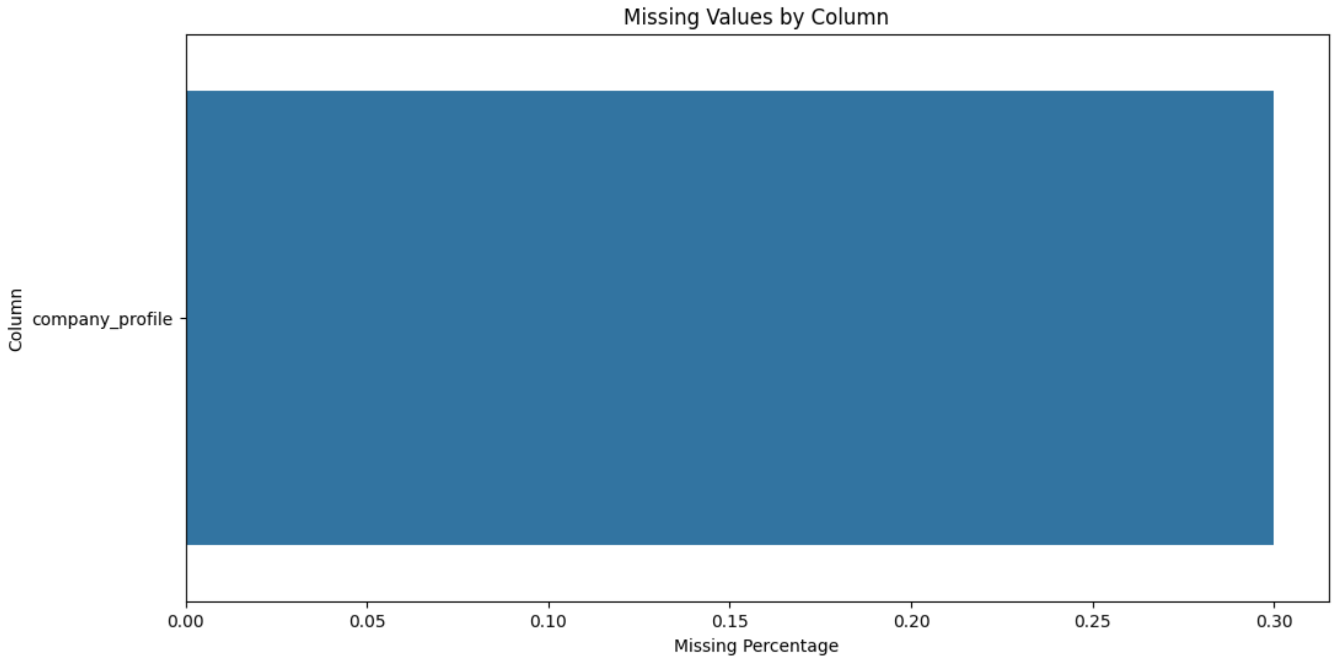
dtype: object

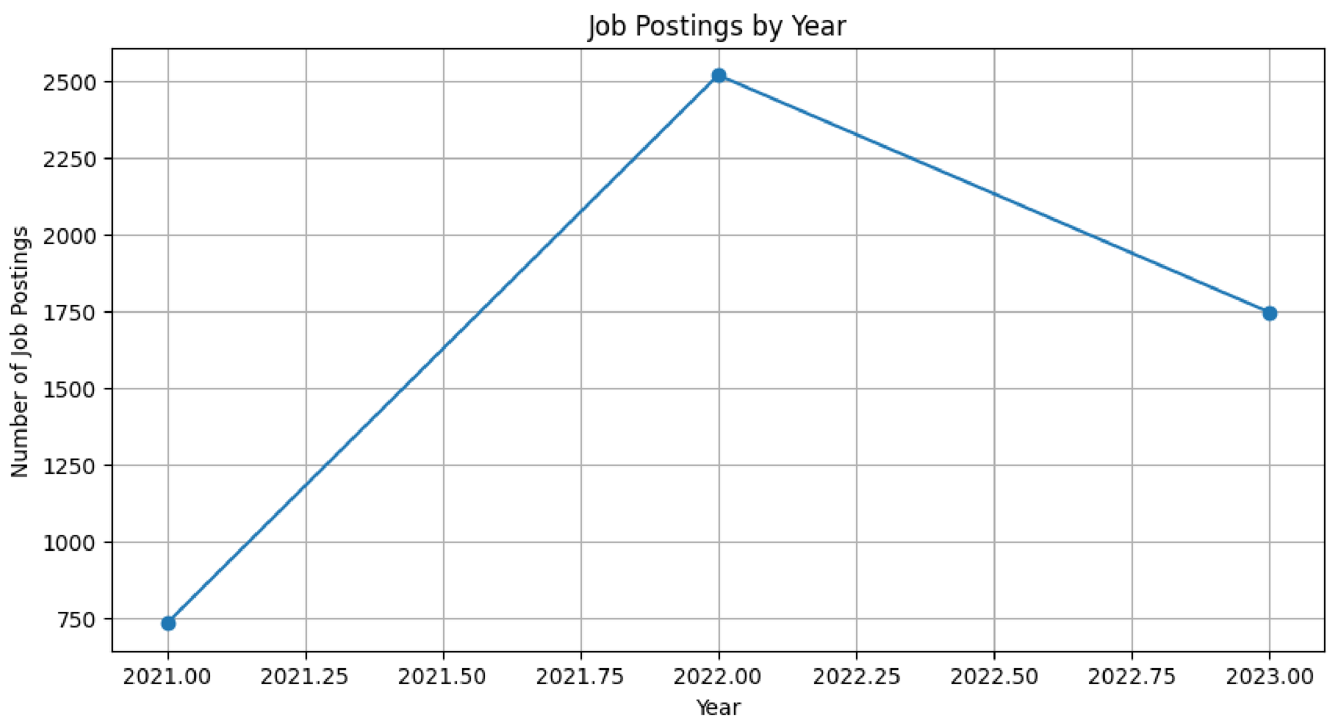
| | dtype | missing_count | missing_% | unique_values |
|---|---|---|---|---|
| company_profile | object | 15 | 0.3 | 882 |
| experience | object | 0 | 0.0 | 48 |
| job_id | float64 | 0 | 0.0 | 4967 |
| salary_range | object | 0 | 0.0 | 560 |
| location | object | 0 | 0.0 | 214 |
| country | object | 0 | 0.0 | 216 |
| qualifications | object | 0 | 0.0 | 10 |
| latitude | float64 | 0 | 0.0 | 216 |
| longitude | float64 | 0 | 0.0 | 216 |
| company_size | int64 | 0 | 0.0 | 4900 |
| work_type | object | 0 | 0.0 | 5 |
| preference | object | 0 | 0.0 | 3 |
| contact_person | object | 0 | 0.0 | 4813 |
| contact | object | 0 | 0.0 | 4998 |
| job_posting_date | object | 0 | 0.0 | 730 |
| job_title | object | 0 | 0.0 | 147 |
| role | object | 0 | 0.0 | 376 |
| job_description | object | 0 | 0.0 | 376 |
| job_portal | object | 0 | 0.0 | 16 |
| benefits | object | 0 | 0.0 | 11 |
| skills | object | 0 | 0.0 | 376 |
| responsibilities | object | 0 | 0.0 | 375 |
| company | object | 0 | 0.0 | 886 |

| | count |
|---|---|
| posting_year | |
| 2021 | 732 |
| 2022 | 2521 |
| 2023 | 1747 |

dtype: int64

MISSING VALUE VISUALIZATION



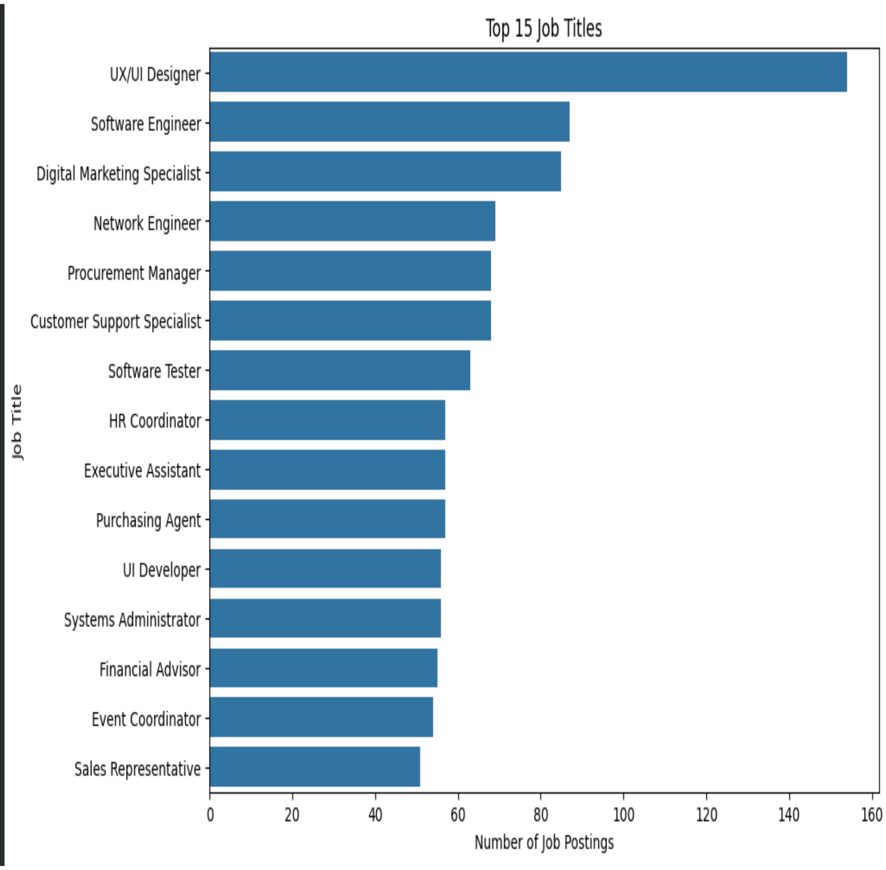



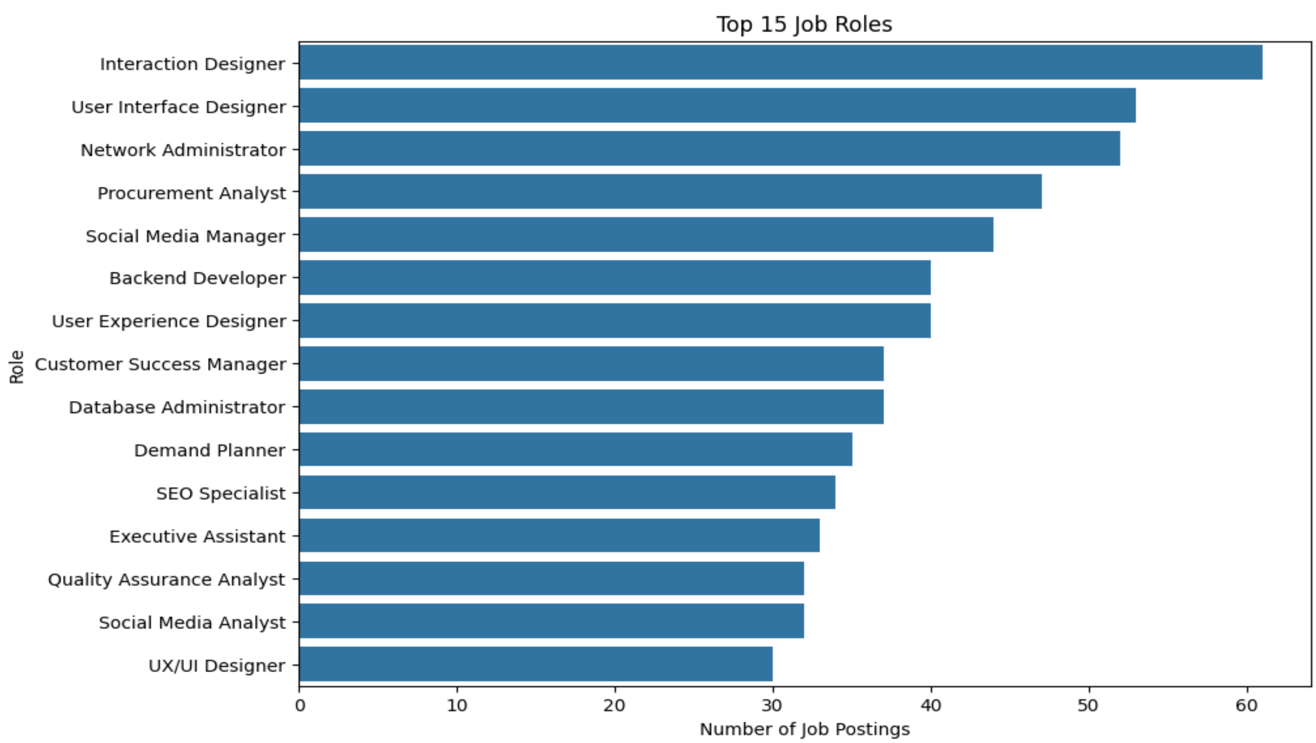
JOB VS COUNT

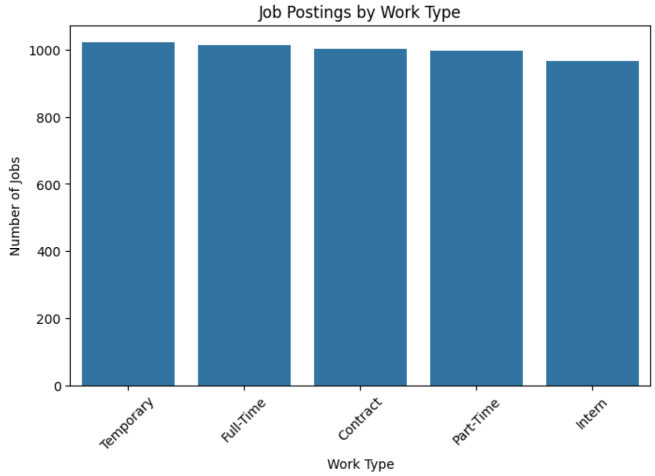
| job_title | count |
|---|---|
| UX/UI Designer | 154 |
| Software Engineer | 87 |
| Digital Marketing Specialist | 85 |
| Network Engineer | 69 |
| Procurement Manager | 68 |
| Customer Support Specialist | 68 |
| Software Tester | 63 |
| HR Coordinator | 57 |
| Executive Assistant | 57 |
| Purchasing Agent | 57 |
| UI Developer | 56 |
| Systems Administrator | 56 |
| Financial Advisor | 55 |
| Event Coordinator | 54 |
| Sales Representative | 51 |

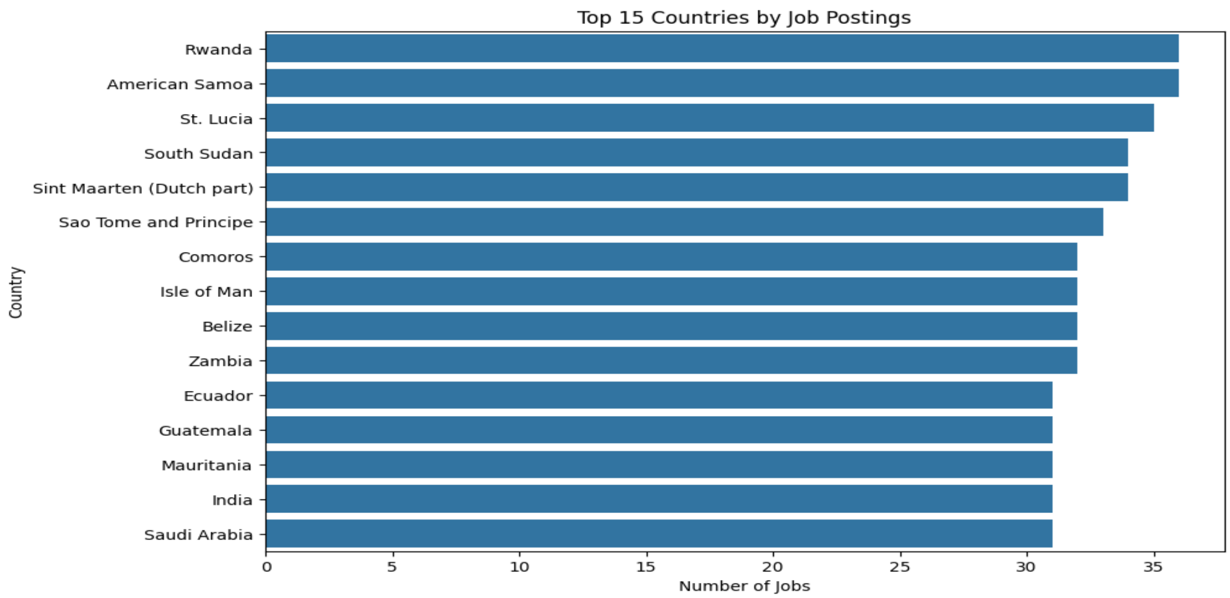
dtype: int64

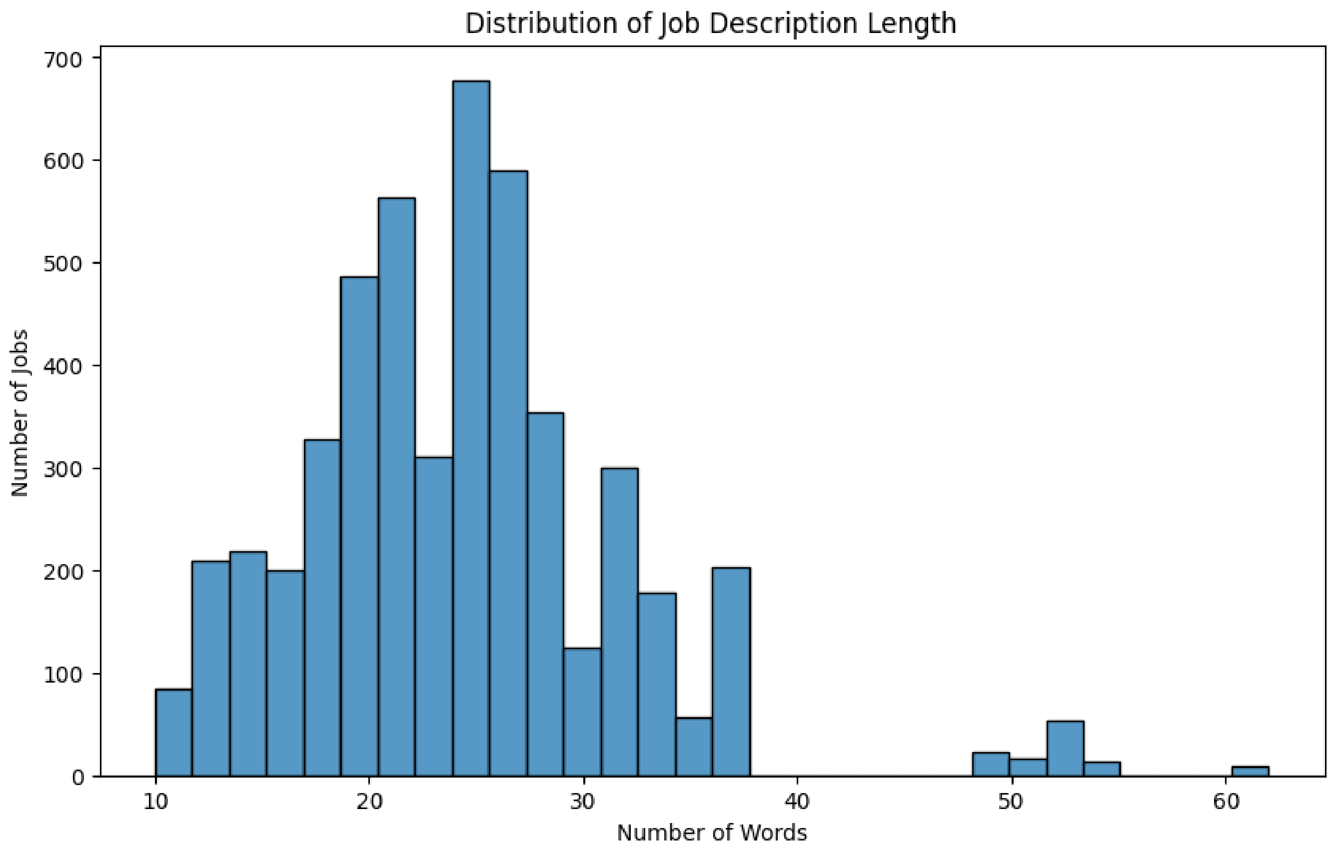
TOP JOB TITLES




Job Postings by Work Type
Number of Jobs
0
200
400
600
800
1000
Temporary
Full-Time
Contract
Part-Time
Intern
Work Type


Top 15 Countries by Job Postings
Country
Rwanda
American Samoa
St. Lucia
South Sudan
Sint Maarten (Dutch part)
Sao Tome and Principe
Comoros
Isle of Man
Belize
Zambia
Ecuador
Guatemala
Mauritania
India
Saudi Arabia
0
5
10
15
20
25
30
35
Number of Jobs




## DAY 4 — Data Cleaning

### Objective

To clean raw job description text so it is suitable for downstream NLP processing — a foundational step for any text-based extraction pipeline.

### Work Completed

- Removed HTML tags from job description text.
- Stripped URLs and email addresses.
- Removed phone numbers and unnecessary special characters/symbols.
- Normalized whitespace (collapsed extra spaces) and removed duplicate text fragments.
- Lower-cased and trimmed the cleaned text.
- Applied the cleaning function across the full dataset to produce a new clean_description column.

### Cleaning Function

```
import re
def clean_text(text):
    text = str(text)
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"<.*?>", "", text)
    text = re.sub(r"\S+@\S+", "", text)
    text = re.sub(r"[^a-zA-Z0-9\s]", " ", text)
    text = re.sub(r"\s+", " ", text)
    return text.strip().lower()
```

```
df["clean_description"] = df["job_description"].apply(clean_text)
```

### Tools & Technologies Used

Python, Pandas, re (regex)

### Outcome

A normalized clean_description column, free of HTML, links, emails, phone numbers, and noisy symbols, forming the basis for tokenization and skill extraction.

### Deliverable

**clean_jobs.csv**

## DAY 5 — Skill Taxonomy Creation

### Objective

To build a master, categorized dictionary of skills — the single most important reference artifact for the extraction logic used in later stages.

### Work Completed

- Compiled a master list of skills spanning programming, databases, analytics tools, machine learning, and cloud platforms (e.g. Python, SQL, Java, C++, Power BI, Tableau, Excel, AWS, Azure, GCP, TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Spark, Hadoop, Docker, Kubernetes, Git).
- Grouped skills into categories: Programming, Database, Data Analytics, Machine Learning, Cloud, and others as needed.
- Defined alias/synonym mappings so variant spellings resolve to one canonical skill name (e.g. python3 -> Python, ms excel -> Excel, postgres/postgresql -> PostgreSQL, powerbi -> Power BI, scikit learn -> Scikit-learn).
- Structured the taxonomy as a table with skill, category, and aliases columns for direct use by the extractor.

### Taxonomy Structure

```
skill      category     aliases
Python     Programming  python3
SQL        Database     sql
Power BI   BI           powerbi, power-bi
PostgreSQL Database     postgres, postgresql
```

### Tools & Technologies Used

Python, Pandas, CSV

### Outcome

A reusable master skill dictionary with categories and alias mappings, enabling consistent, normalized skill matching regardless of how a skill is phrased in a job posting.

```python
# ============================================================
# JOB SKILL TAXONOMY + SYNONYMS / ALIASES
# ============================================================
skill_taxonomy = {

    # ============================================================
    # 1. PROGRAMMING
    # ============================================================
    "Programming": {

        "Python": [
            "python",
            "python programming",
            "python development",
            "python scripting",
            "python language"
        ],

        "Java": [
            "java",
            "java programming",
            "java development",
            "java language"
        ],

        "C": [
            "c",
            "c programming",
            "c language"
        ],

        "C++": [
            "c++",
            "cpp",
            "c plus plus",
            "c++ programming"
        ],
```

| | | | |
|---|---|---|---|
| 1 | Programming | Python | python programming |
| 2 | Programming | Python | python development |
| 3 | Programming | Python | python scripting |
| 4 | Programming | Python | python language |
| 5 | Programming | Java | java |
| 6 | Programming | Java | java programming |
| 7 | Programming | Java | java development |
| 8 | Programming | Java | java language |
| 9 | Programming | C | c |
| 10 | Programming | C | c programming |
| 11 | Programming | C | c language |
| 12 | Programming | C++ | c++ |
| 13 | Programming | C++ | cpp |
| 14 | Programming | C++ | c plus plus |
| 15 | Programming | C++ | c++ programming |
| 16 | Programming | C# | c# |
| 17 | Programming | C# | c sharp |
| 18 | Programming | C# | csharp |
| 19 | Programming | C# | c# programming |

```
Duplicate Synonyms:
```

| | category | canonical_skill | synonym |
|---|---|---|---|
| 790 | Finance & Accounting | Account Management | account executive |
| 590 | Sales | Account Management | account executive |
| 784 | Finance & Accounting | Account Management | account management |
| 584 | Sales | Account Management | account management |
| 787 | Finance & Accounting | Account Management | account manager |
| ... | ... | ... | ... |
| 902 | Software Testing & QA | Automated Testing | test automation |
| 1038 | Event Management | Vendor Management | vendor management |
| 1105 | Supply Chain & Logistics | Vendor Management | vendor management |
| 741 | UI/UX Design | Graphic Design | visual design |
| 720 | UI/UX Design | Visual Design | visual design |

84 rows × 3 columns

```
TEST RESULTS:

python                          → {'canonical_skill': 'Python', 'category': 'Programming'}
sklearn                         → {'canonical_skill': 'Scikit-learn', 'category': 'Machine Learning'}
mysql                           → {'canonical_skill': 'MySQL', 'category': 'Database'}
amazon web services             → {'canonical_skill': 'AWS', 'category': 'Cloud'}
aws                             → {'canonical_skill': 'AWS', 'category': 'Cloud'}
pytorch                         → {'canonical_skill': 'PyTorch', 'category': 'Machine Learning'}
xg boost                        → {'canonical_skill': 'XGBoost', 'category': 'Machine Learning'}
powerbi                         → {'canonical_skill': 'Power BI', 'category': 'Business Intelligence'}
gcp                             → {'canonical_skill': 'Google Cloud', 'category': 'Cloud'}
k8s                             → {'canonical_skill': 'Kubernetes', 'category': 'DevOps'}
reactjssocial media             → None
```

### Deliverable

**skill_taxonomy.csv**

## DAY 6 — NLP Text Preprocessing

### Objective

To convert cleaned job description text into a linguistically normalized form suitable for reliable skill matching.

### Work Completed

- Performed sentence segmentation and word tokenization on the cleaned text.
- Lower-cased all tokens for case-insensitive matching.
- Removed stopwords (e.g. we, are, for, with) to retain only meaningful terms.
- Applied lemmatization to reduce words to their dictionary base form.
- Ran a stemming comparison alongside lemmatization to evaluate which normalization method preserves skill-name integrity better (important since stemming can distort technical terms).

### Example Transformation

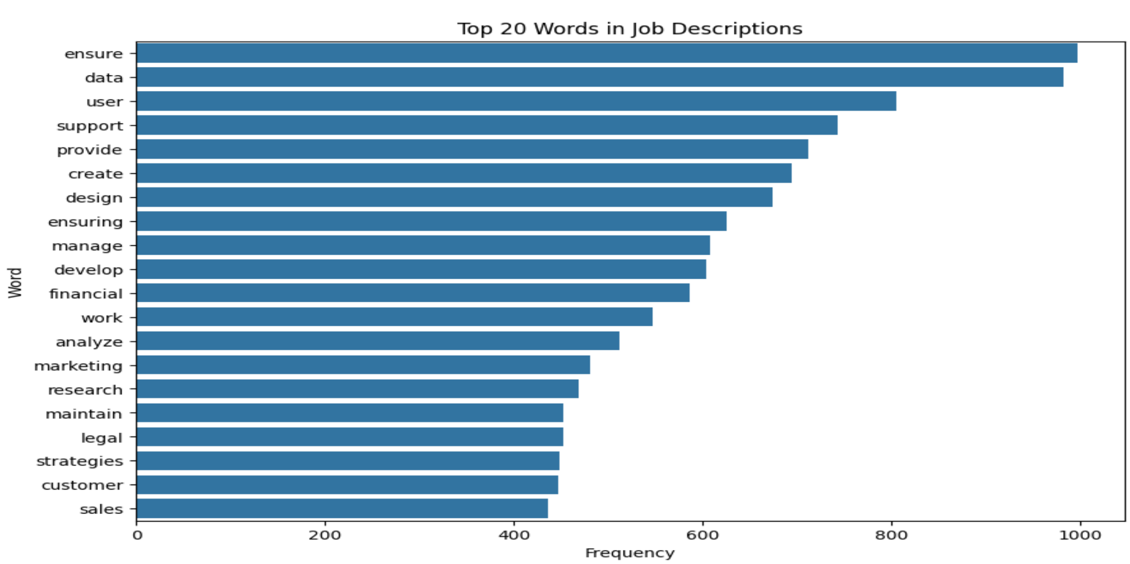
```
Original:
We are looking for candidates with strong experience in Python and SQL.

Tokenized:
we | are | looking | for | candidates | with | strong | experience | in | python | and | sql

After stopword removal:
looking | candidates | strong | experience | python | sql
```

### Tools & Technologies Used

Python, NLTK, spaCy, Jupyter Notebook

### Outcome

A preprocessed, tokenized version of every job description, with stopwords removed and terms normalized, ready for dictionary-based skill matching.



```
#tokenization
tokens = word_tokenize(lower_text)

print("Tokens:")
print(tokens)
```

```
Tokens:
['we', 'are', 'looking', 'for', 'candidates', 'with', 'strong', 'experience', 'in', 'python', 'and', 'sql', '.', 'the', 'candidate', 'should', 'have', 'good', 'knowledge', 'of', 'power', 'bi', 'and', 'aws', '.']
```

AFTER STOPWORD REMOVAL

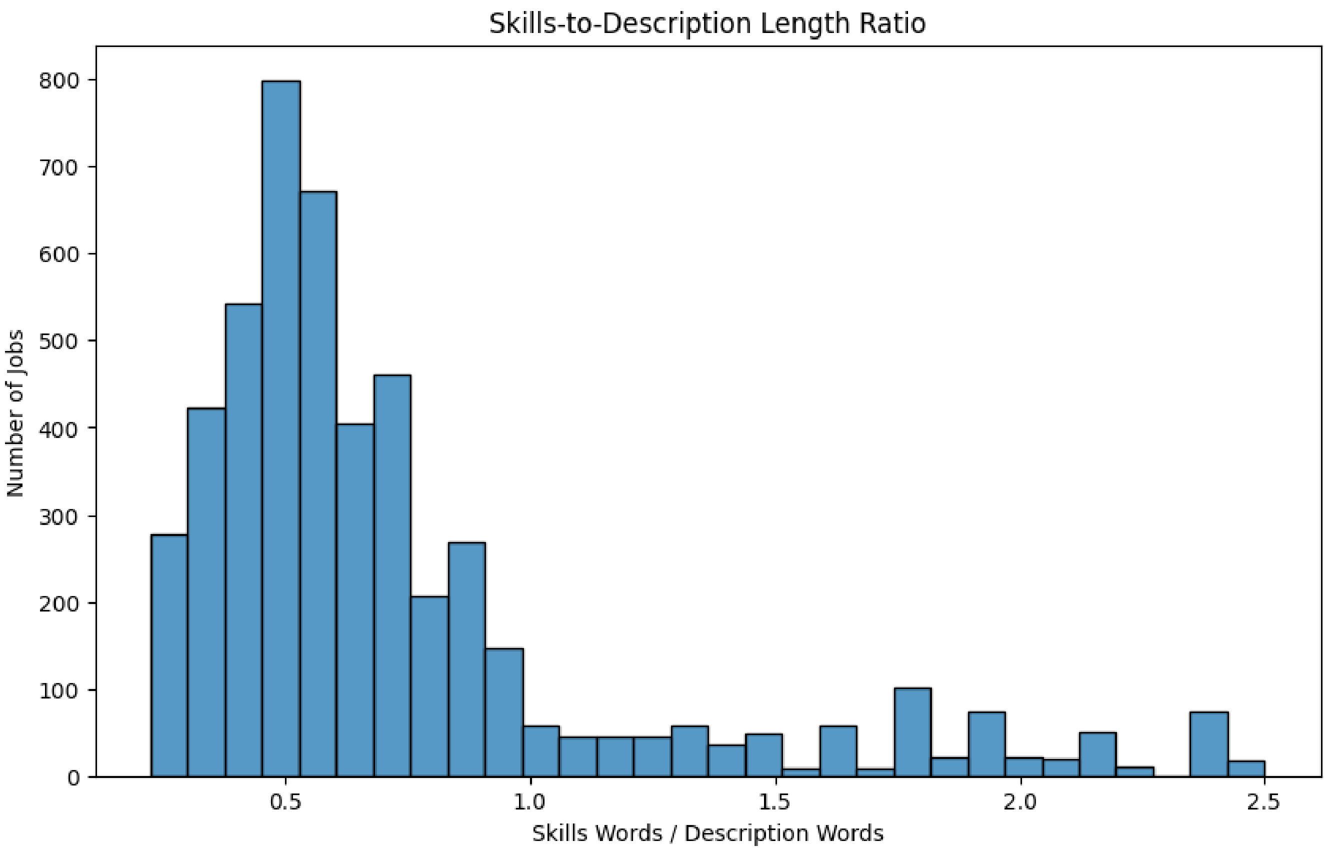
```
After Stopword Removal:
['looking', 'candidates', 'strong', 'experience', 'python', 'sql', 'candidate', 'good', 'knowledge', 'power', 'bi', 'aws']
```

## Deliverable

**NLP_Preprocessing.ipynb**

WORD CLOUD VISUALIZATION


Job Description Word Cloud
products services
user experience
decision making
social media
visually appealing
user interface
supply chain
managers oversee
digital interfaces
user friendly search engine
web application
designers create
customer satisfaction
design implement
within organization
provide insights
administrators manage
computer networks
ensure product
development teams
provide legal
data provide
quality standards
ensure compliance
analyze social
analyze data
ensuring data
saving opportunities
market trends
user interactions
digital marketing
cost saving
insights support
data driven
data integrity
medical care
brand awareness
plan execute
ensure efficient
develop execute
quality assurance
create visually
negotiate contracts


Skills-to-Description Length Ratio
Number of Jobs
0
100
200
300
400
500
600
700
800
0.5
1.0
1.5
2.0
2.5
Skills Words / Description Words

## DAY 7 — Rule-Based Skill Extractor

### Objective

To build the first working version of the skill extraction engine using direct dictionary lookups against the Day 5 taxonomy.

### Work Completed

- Implemented a dictionary-based matching function that scans each cleaned job description for every skill in the master skill list.
- For each job description, searched for occurrences of each known skill (e.g. Python, SQL, Power BI, Tableau, Excel, AWS) using case-insensitive substring matching.
- Collected all matched skills per job description into a result list.
- Validated the extractor against sample descriptions to confirm expected skills were correctly identified.

### Extraction Function

```
skills = ["python", "sql", "power bi", "tableau", "excel", "aws"]

def extract_skills(text):
    found = []
    for skill in skills:
        if skill in text.lower():
            found.append(skill)
    return found

# Sample output: Python, SQL, Power BI, AWS
```

### Tools & Technologies Used

Python, Pandas

### Outcome

A functioning rule-based skill extractor that reliably tags known skills from job descriptions, serving as the baseline before more advanced (NER/ML-based) extraction methods.

### Deliverable

**rule_based_extractor.py**

## Summary & Next Steps

Across Days 1–7, the project moved from problem definition and dataset collection through exploratory analysis, text cleaning, taxonomy design, NLP preprocessing, and a first working rule-based skill extractor. The pipeline currently extracts skills via exact/substring dictionary matching against the Day 5 taxonomy.

# DAY 8 — Improve Skill Extraction Using Regex & Phrase Matching

**1. Objective**

The objective of Day 8 was to improve the **rule-based skill extraction system** developed on Day 7.

The basic dictionary-based approach can have problems with variations of the same skill. For example:

- Python
- Python 3
- Python3
- Python programming

All of these should ideally be mapped to the same canonical skill: Python

Therefore, regular expressions and phrase matching were introduced to improve skill recognition.

## 2. Problem With Basic Dictionary Matching

The initial extractor searched for skills directly in the job description.

For example:

```
skills = [
    "python",
    "sql",
    "power bi",
    "tableau",
    "excel",
    "aws"
]
```

A simple implementation was:

```
def extract_skills(text):
    found = []

    for skill in skills:
        if skill in text.lower():
            found.append(skill)
```

```
    return found
```

Although this works for straightforward cases, it has limitations.

**Example**

A job description may contain: Python 3 programming experience

A simple search for "python" can identify Python, but different variations such as:

Python3

Python programming

Python 3

need to be handled consistently.

There can also be problems with:

- word boundaries
- different spellings
- multi-word skills
- abbreviations
- skill aliases
- duplicate matches

## 3. Regular Expression Matching

Regular expressions were introduced to identify skill variations more reliably.

Example: import re

```
pattern = r"\bpython(?:\s*3)?\b"
```

This pattern can recognize variations such as:

Python

Python 3

Python3

**Explanation**

\b represents a word boundary. Python matches the word Python.

(?:\s*3)? optionally matches:

- 3
- 3
- nothing

Therefore:

Python

Python 3

Python3

can all be recognized as Python.

## 4. Why Phrase Matching Was Needed

Job descriptions frequently contain multi-word skills.

Examples from the taxonomy include:

machine learning

deep learning

natural language processing

power bi

data science

social media

account management

data visualization

Searching for individual words is not sufficient.

For example:

machine learning

should be recognized as **one skill**, rather than treating:

machine

learning

as separate skills.

Therefore, phrase matching was implemented.

```
[{'category': 'Data Science',
  'canonical_skill': 'Data Science',
  'matched_synonym': 'data scientist'},
 {'category': 'Programming',
  'canonical_skill': 'Python',
  'matched_synonym': 'python'},
 {'category': 'Machine Learning',
  'canonical_skill': 'Scikit-learn',
  'matched_synonym': 'scikit-learn'},
 {'category': 'Machine Learning',
  'canonical_skill': 'XGBoost',
  'matched_synonym': 'xgboost'},
 {'category': 'Database', 'canonical_skill': 'SQL', 'matched_synonym': 'sql'},
 {'category': 'Database',
  'canonical_skill': 'MySQL',
  'matched_synonym': 'mysql'},
 {'category': 'Cloud', 'canonical_skill': 'AWS', 'matched_synonym': 'aws'}]
```

JSON FORMAT JOB TITLE VS SKILLS EXTRACTED

[{"index":0,"job_title":"Digital Marketing Specialist","extracted_skills":"{'category': 'Digital Marketing', 'canonical_skill': 'Social Media', 'matched_synonym': 'social media'},{'category': 'Digital Marketing', 'canonical_skill': 'Brand Marketing', 'matched_synonym': 'brand awareness'}"},{"index":1,"job_title":"Web Developer","extracted_skills":"{'category': 'Web Development', 'canonical_skill': 'Web Development', 'matched_synonym': 'web developers'}"},{"index":2,"job_title":"Operations Manager","extracted_skills":"{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality control'},{'category': 'Operations', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality control'}"},{"index":3,"job_title":"Network Engineer","extracted_skills":"{'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym': 'wireless network'}"},{"index":4,"job_title":"Event Manager","extracted_skills":"{'category': 'Event Management', 'canonical_skill': 'Logistics', 'matched_synonym': 'logistics'},{'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_synonym': 'budgeting'}"},{"index":5,"job_title":"Software Tester","extracted_skills":"{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Assurance', 'matched_synonym': 'quality assurance'},{'category': 'Operations', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality assurance'}"},{"index":6,"job_title":"Teacher","extracted_skills":"{'category': 'Education & Teaching', 'canonical_skill': 'Teaching', 'matched_synonym': 'teacher'},{'category': 'Education & Teaching', 'canonical_skill': 'Lesson Planning', 'matched_synonym': 'lesson plans'}"},{"index":7,"job_title":"UX/UI Designer","extracted_skills":"{'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'user interface designers'}"},{"index":8,"job_title":"UX/UI Designer","extracted_skills":"{'category': 'UI/UX Design', 'canonical_skill': 'UX Design', 'matched_synonym': 'interaction designers'},{'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'digital interfaces'}"},{"index":9,"job_title":"Wedding Planner","extracted_skills":"{'category': 'Event Management', 'canonical_skill': 'Budget Management', 'matched_synonym': 'budget management'},{'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_synonym': 'budget management'}"}]

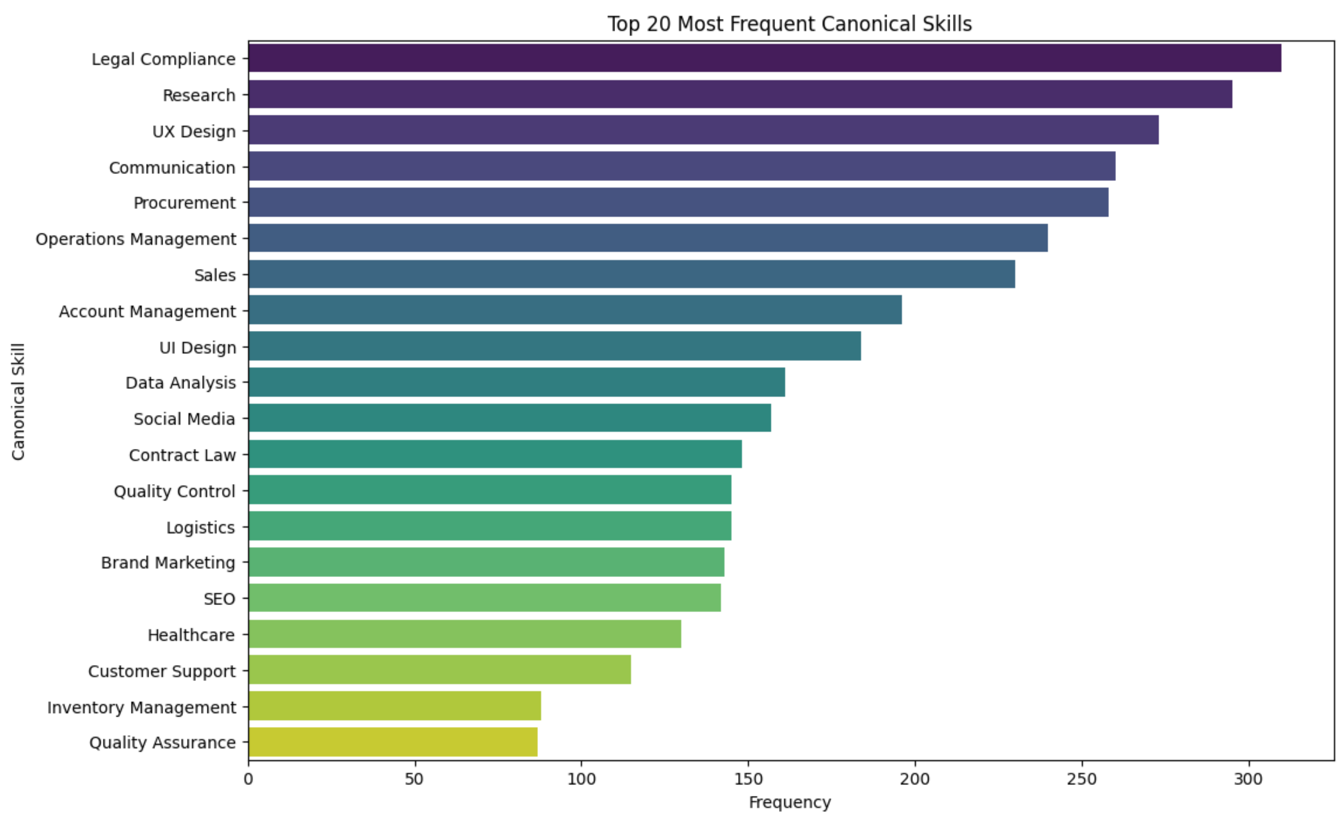
CSV FORMAT OF IMPROVED PARSE AND SKILL MATCHING

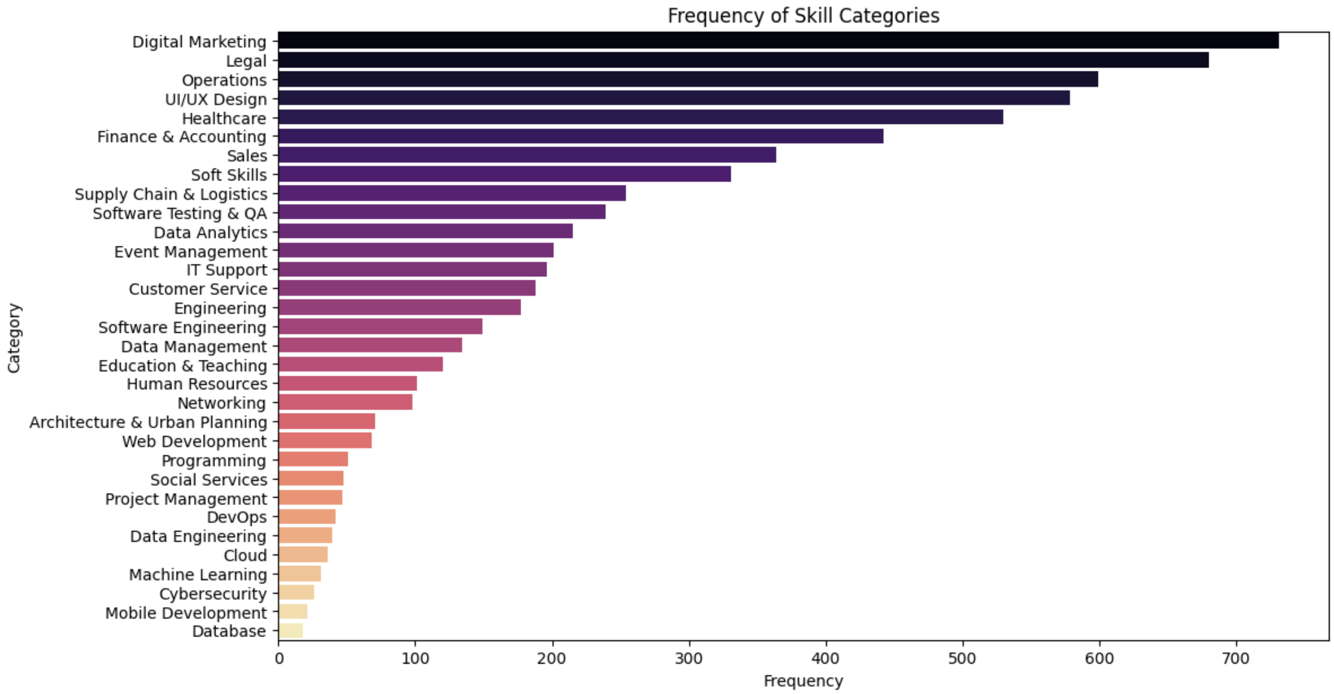
index,job_index,job_id,job_title,category,canonical_skill,matched_synonym

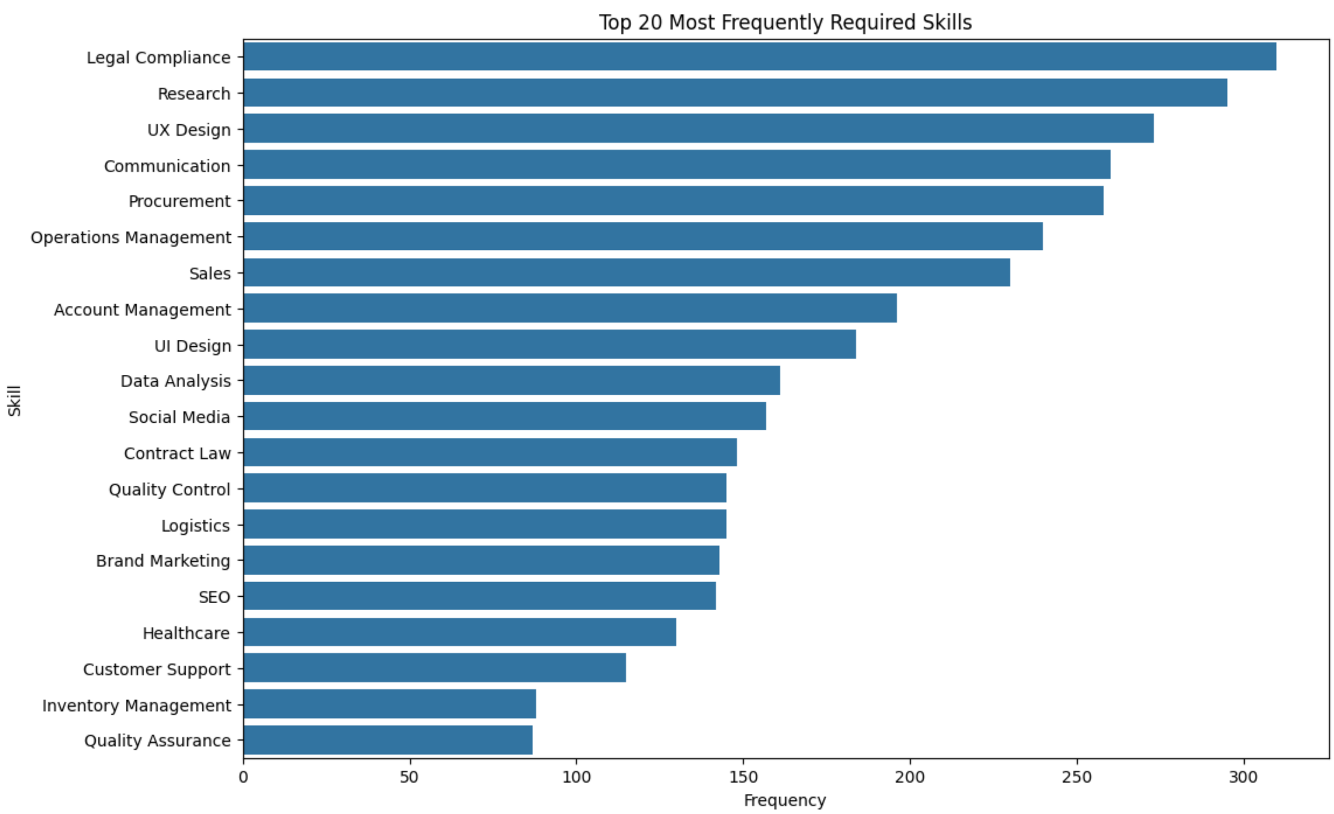
0,0,1089840000000000.0,Digital Marketing Specialist,Digital Marketing,Social Media,social media

1,0,1089840000000000.0,Digital Marketing Specialist,Digital Marketing,Brand Marketing,brand awareness

2,1,398454000000000.0,Web Developer,Web Development,Web Development,web developers
3,2,481640000000000.0,Operations Manager,Software Testing & QA,Quality Control,quality control
4,2,481640000000000.0,Operations Manager,Operations,Quality Control,quality control
5,3,688193000000000.0,Network Engineer,Networking,Wireless Networking,wireless network
6,4,117058000000000.0,Event Manager,Event Management,Logistics,logistics
7,4,117058000000000.0,Event Manager,Finance & Accounting,Budgeting,budgeting
8,5,116831000000000.0,Software Tester,Software Testing & QA,Quality Assurance,quality assurance
9,5,116831000000000.0,Software Tester,Operations,Quality Control,quality assurance
10,6,1292170000000000.0,Teacher,Education & Teaching,Teaching,teacher
11,6,1292170000000000.0,Teacher,Education & Teaching,Lesson Planning,lesson plans
12,7,1498780000000000.0,UX/UI Designer,UI/UX Design,UI Design,user interface designers
13,8,1680290000000000.0,UX/UI Designer,UI/UX Design,UX Design,interaction designers
14,8,1680290000000000.0,UX/UI Designer,UI/UX Design,UI Design,digital interfaces
15,9,255628000000000.0,Wedding Planner,Event Management,Budget Management,budget management
16,9,255628000000000.0,Wedding Planner,Finance & Accounting,Budgeting,budget management
17,10,2696960000000000.0,QA Analyst,Software Testing & QA,Performance Testing,performance testing
18,11,1446190000000000.0,Litigation Attorney,Legal,Family Law,family law
19,12,1914120000000000.0,Mechanical Engineer,Engineering,Mechanical Engineering,mechanical design






## 5. spaCy PhraseMatcher

The project uses **spaCy PhraseMatcher** to identify skill phrases inside job descriptions.

The general process is:

Job Description
↓
spaCy NLP Processing
↓
PhraseMatcher
↓
Skill Phrase Detection
↓
Synonym Mapping
↓
Canonical Skill
↓
Skill Category

## 6. Taxonomy-Based Phrase Matching

Instead of maintaining only a small list of skills, the project uses the larger **skill taxonomy** developed during the previous stages.

The taxonomy contains:

Category
↓
Canonical Skill
↓
Synonyms / Aliases

**For example:**

```
Programming
   Python
       python
       python programming
       python development
       python scripting
```

This allows different expressions to map to the same canonical skill.

## 7. Synonym-to-Skill Mapping

A mapping was created so that detected synonyms can be converted into their canonical skills.

Example:

synonym_to_skill["python"]  maps to:

canonical_skill → Python

category → Programming

**Similarly:**

Sklearn maps to:

Scikit-learn

Machine Learning

and:

powerbi maps to:

Power BI

Business Intelligence

This is important because the final output should contain standardized skills instead of every variation found in the text.

## 8. PhraseMatcher Implementation

The taxonomy synonyms were converted into spaCy patterns.

Conceptually:

```
matcher = PhraseMatcher(nlp.vocab, attr="LOWER")
```

Using:

```
attr="LOWER"
```

allows matching without depending on capitalization.

**For example:**

- Python
- python
- PYTHON

can be treated consistently.

Patterns are then created from the synonyms:

```
patterns = []

for synonym in synonym_to_skill.keys():
    patterns.append(nlp.make_doc(synonym))
```

The patterns are added to the matcher:

```
matcher.add("SKILLS", patterns)
```

## 9. Skill Extraction Function

A spaCy-based extraction function was developed to process each job description.

The function performs the following operations:

**Input Job Description**

↓

**Convert text into spaCy Doc**

↓

**Run PhraseMatcher**

**↓**

**Find matching skill phrases**

**↓**

**Look up synonym in taxonomy**

**↓**

**Get canonical skill**

**↓**

**Get category**

**↓**

**Remove duplicate canonical skills**

**↓**

**Return structured result**

The resulting information contains fields such as:

**category**

**canonical_skill**

**matched_synonym**

**Example:**

**Python → Programming → matched synonym: python**

### 10. Duplicate Skill Handling

One important improvement was duplicate removal.

For example, a job description might contain:

- Python
- Python programming
- Python

The extractor should not produce:

- Python
- Python
- Python programming

Instead, all of these should map to: Python

Therefore, the final extraction uses the canonical skill rather than simply returning every phrase match.

## 11. Testing the Extractor

A test job description was used containing several skills:

- Python
- Python programming
- SQL

- Power BI
- machine learning
- natural language processing

The raw PhraseMatcher output initially showed the individual matched phrases.

After applying canonical mapping and duplicate removal, the output became:

- Python
- SQL
- Power BI
- Machine Learning
- Natural Language Processing

This demonstrates that different mentions can be normalized into unique canonical skills.

| | job_index | job_id | job_title | category | canonical_skill | matched_synonym |
|---|---|---|---|---|---|---|
| 0 | 0 | 1.089840e+15 | Digital Marketing Specialist | Digital Marketing | Social Media | social media |
| 1 | 0 | 1.089840e+15 | Digital Marketing Specialist | Digital Marketing | Brand Marketing | brand awareness |
| 2 | 1 | 3.984540e+14 | Web Developer | Web Development | Web Development | web developers |
| 3 | 2 | 4.816400e+14 | Operations Manager | Software Testing & QA | Quality Control | quality control |
| 4 | 2 | 4.816400e+14 | Operations Manager | Operations | Quality Control | quality control |
| 5 | 3 | 6.881930e+14 | Network Engineer | Networking | Wireless Networking | wireless network |
| 6 | 4 | 1.170580e+14 | Event Manager | Event Management | Logistics | logistics |
| 7 | 4 | 1.170580e+14 | Event Manager | Finance & Accounting | Budgeting | budgeting |
| 8 | 5 | 1.168310e+14 | Software Tester | Software Testing & QA | Quality Assurance | quality assurance |
| 9 | 5 | 1.168310e+14 | Software Tester | Operations | Quality Control | quality assurance |
| 10 | 6 | 1.292170e+15 | Teacher | Education & Teaching | Teaching | teacher |
| 11 | 6 | 1.292170e+15 | Teacher | Education & Teaching | Lesson Planning | lesson plans |
| 12 | 7 | 1.498780e+15 | UX/UI Designer | UI/UX Design | UI Design | user interface designers |
| 13 | 8 | 1.680290e+15 | UX/UI Designer | UI/UX Design | UX Design | interaction designers |
| 14 | 8 | 1.680290e+15 | UX/UI Designer | UI/UX Design | UI Design | digital interfaces |
| 15 | 9 | 2.556280e+14 | Wedding Planner | Event Management | Budget Management | budget management |
| 16 | 9 | 2.556280e+14 | Wedding Planner | Finance & Accounting | Budgeting | budget management |
| 17 | 10 | 2.696960e+15 | QA Analyst | Software Testing & QA | Performance Testing | performance testing |
| 18 | 11 | 1.446190e+15 | Litigation Attorney | Legal | Family Law | family law |
| 19 | 12 | 1.914120e+15 | Mechanical Engineer | Engineering | Mechanical Engineering | mechanical design |

## 13. Applying the Extractor to the 5K Dataset

After testing the extractor, it was applied to the 5,000-job dataset.

The extraction was performed on:

df["job_description_matching"]

and the results were stored in:

df["extracted_skills_spacy"]

This created a structured skill-extraction result for each job description.

### 14. Initial Results

After applying the improved extraction approach to the 5K dataset, approximately:

- Total Jobs: 5,000
- Jobs with detected skills: ~3,650
- Jobs without detected skills: ~1,350
- Skill coverage: ~72%

After Improvement :

```
#Checking how many job actually received skills
jobs_with_skills = df["extracted_skills"].apply(len).gt(0).sum()
total_jobs = len(df)

coverage = (jobs_with_skills / total_jobs) * 100

print("Total Jobs:", total_jobs)
print("Jobs with at least one skill:", jobs_with_skills)
print("Skill Extraction Coverage:", round(coverage, 2), "%")
```

```
Total Jobs: 5000
Jobs with at least one skill: 3835
Skill Extraction Coverage: 76.7 %
```

```
jobs_without_skills = total_jobs - jobs_with_skills

print("Jobs without extracted skills:", jobs_without_skills)
```

```
Jobs without extracted skills: 1165
```

### 15. Analysis of Zero-Skill Jobs

The zero-skill jobs were then inspected to understand whether the issue was with the extractor or the taxonomy.

Examples included:

- **Key Account Manager**

strategic account managers nurture relationships with key clients...

**Potential missing concepts:**

Account Management

Account Strategy

Client Relationship Management

- **Data Engineer**

a data architect designs and manages data infrastructure...

**Potential missing concepts:**

Data Architecture

Data Infrastructure

- **Technical Writer**

documentation specialists create and manage documents...

**Potential missing concepts:**

Technical Writing

Documentation

This analysis showed that some zero-skill results were caused by **missing taxonomy synonyms**, rather than a failure of PhraseMatcher.

## 16. Important Finding

Not every zero-skill job represents an extraction failure.

Some jobs belong to domains that are currently outside the project's taxonomy.

**Examples observed include:**

- Nurse Practitioner
- Wedding Planner
- Urban Planner
- Structural Engineer
- Investment Analyst

Therefore, forcing these jobs into the existing technology/data/business taxonomy could create **false-positive skill extraction**.

The objective is therefore not simply:

Zero skills = 0

Instead, the goal is:

**Maximize meaningful skill extraction while minimizing false positives.**

## 17. Day 8 Outcome

By the end of Day 8, the skill extraction system was improved from simple dictionary matching to a more robust approach using:

- Regular expressions
- spaCy NLP
- spaCy PhraseMatcher
- Multi-word phrase matching
- Case-insensitive matching
- Skill synonyms and aliases
- Canonical skill mapping
- Duplicate canonical-skill removal
- Taxonomy-based categorization
- Zero-skill analysis

**Final pipeline**

5K Job Dataset

↓

Job Description

↓

Text Preprocessing

↓

Regex / Phrase Matching

↓

spaCy PhraseMatcher

↓

Synonym Detection

↓

Canonical Skill Mapping

↓

Category Mapping

↓

Duplicate Removal

↓

Structured Extracted Skills

↓

Skill Analysis

## 18. Deliverable

**Deliverable: NLP_Skill_Extraction_Day8.ipynb**

The notebook contains:

1. Regex-based skill matching
2. spaCy setup
3. PhraseMatcher implementation
4. Taxonomy integration
5. Synonym mapping
6. Canonical skill normalization
7. Duplicate removal
8. Test cases
9. Extraction on the 5K dataset
10. Skill coverage analysis
11. Zero Skills Jobs analysis

## Day 9 — spaCy Phrase Matching & Skill Extraction Documentation

### Objective

Improve the rule-based skill extraction system by using spaCy PhraseMatcher to identify skills and their variations from job descriptions.

### Work Performed

- Loaded the en_core_web_sm spaCy model.
- Created a PhraseMatcher using case-insensitive matching.
- Used the existing skill taxonomy and synonyms/aliases to create matching patterns.
- Matched skills against the job_description_matching column.
- Mapped matched phrases to their canonical skill and category.
- Removed duplicate skill matches.
- Stored the extracted results in a new column:

extracted_skills_spacy

### Example

### Job Description:

Candidates should have experience with Python programming, SQL and Power BI.

Extracted Skills:

- Python
- SQL
- Power BI

### Output Structure

Each extracted skill contains:

category

canonical_skill

matched_text

### Example:

Data Analytics → Data Analysis → data analysis

Programming → Python → python programming

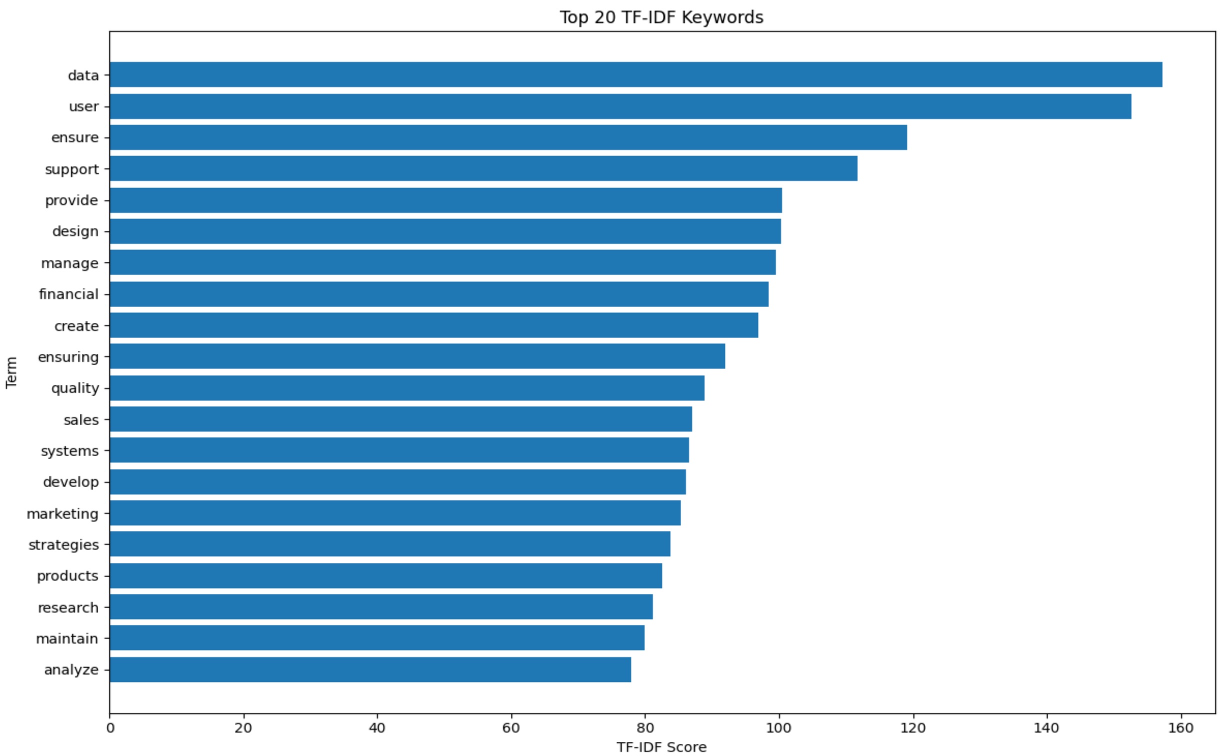
Business Intelligence → Power BI → power bi

## Result

The spaCy PhraseMatcher provided a more flexible approach to matching multi-word phrases and skill synonyms compared with simple string matching. It was applied to the 5,000-record job dataset and stored separately for further comparison and analysis.

```
Number of features: 1000

First 50 features:
['academic' 'access' 'accessibility' 'accessible' 'accordingly' 'account'
 'accountants' 'accounting' 'accounts' 'accuracy' 'accurate' 'accurately'
 'achieve' 'acquiring' 'acquisition' 'acquisitions' 'activities' 'acute'
 'addiction' 'address' 'addressing' 'adhering' 'adjust' 'administer'
 'administers' 'administrative' 'administrators' 'adolescents' 'adoption'
 'adult' 'adults' 'advertising' 'advice' 'advise' 'advisors'
 'aesthetically' 'aesthetics' 'affecting' 'age' 'agencies' 'agent' 'ages'
 'agile' 'agreements' 'aid' 'aided' 'aiding' 'air' 'aircraft' 'align']
```

| | term | tfidf_score |
|---|---|---|
| 0 | data | 157.289586 |
| 1 | user | 152.696132 |
| 2 | ensure | 119.187561 |
| 3 | support | 111.761686 |
| 4 | provide | 100.406095 |
| 5 | design | 100.308497 |
| 6 | manage | 99.482207 |
| 7 | financial | 98.431219 |
| 8 | create | 96.950658 |
| 9 | ensuring | 92.029080 |
| 10 | quality | 88.870584 |




## Tools Used

- Python
- Pandas
- spaCy
- PhraseMatcher

| | | |
|---|---|---|
| 6 | perform | 0.186900 |
| 7 | organization | 0.167091 |
| 8 | standards | 0.164450 |
| 9 | processes | 0.157593 |

```
============================================================
Job: 3
Title: Network Engineer

Top TF-IDF Terms:
```

| | term | tfidf_score |
|---|---|---|
| 0 | wireless | 0.825639 |
| 1 | network | 0.267998 |
| 2 | connectivity | 0.195143 |
| 3 | secure | 0.190791 |
| 4 | communications | 0.188207 |
| 5 | reliable | 0.159088 |
| 6 | troubleshoot | 0.157288 |
| 7 | engineers | 0.119723 |
| 8 | solutions | 0.115590 |
| 9 | optimize | 0.109245 |

```
============================================================
Job: 4
Title: Event Manager

Top TF-IDF Terms:
```

| | term | tfidf_score |
|---|---|---|
| 0 | events | 0.436870 |
| 1 | vendors | 0.282468 |
| 2 | attendees | 0.280649 |
| 3 | catering | 0.271064 |
| 4 | execution | 0.268278 |
| 5 | conferences | 0.245517 |

# DAY 10 — N-Gram Analysis

## 1. Objective

The objective of Day 10 was to identify single-word and multi-word terms from job descriptions using N-Gram analysis.

Many technical and professional skills consist of multiple words, such as:

- Machine Learning
- Deep Learning
- Power BI
- Natural Language Processing
- Data Analysis
- Computer Vision
- Cloud Computing

A simple single-word tokenizer may not capture these skill phrases effectively. Therefore, unigrams, bigrams, and trigrams were generated from the 5,000 job descriptions.

## 2. Dataset Used

- **Dataset:** 5,000 job records
- **Text column:** job_description_matching
- **Task:** Extract important unigrams, bigrams, and trigrams from job descriptions.

## 3. Tools & Libraries

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- TfidfVectorizer

### 4. Methodology

The `TfidfVectorizer` was configured with:

```
TfidfVectorizer(
    ngram_range=(1, 3),
    stop_words="english"
)
```

The configuration generates:

| N-Gram | Meaning | Example |
|---|---|---|
| Unigram | One word | Python |
| Bigram | Two words | Machine Learning |
| Trigram | Three words | Natural Language Processing |

## 5. Implementation Steps

**Step 1 — Prepare Job Descriptions**

Missing descriptions were replaced with empty strings and the text was converted to string format.

**Step 2 — Generate N-Grams**

TF-IDF was applied with:

ngram_range = (1,3)

This generated unigrams, bigrams, and trigrams simultaneously.

**Step 3 — Calculate TF-IDF Scores**

The average TF-IDF score was calculated for each generated term to identify important terms across the dataset.

**Step 4 — Separate N-Grams**

The generated terms were divided into:

- Unigrams
- Bigrams
- Trigrams

**Step 5 — Compare With Skill Taxonomy**

The discovered multi-word terms were compared against the existing **skill taxonomy and synonyms**.

This helped identify terms that may already exist in the taxonomy and potential terms that are not currently included.

## 6. Key Outputs

The following outputs were generated:

**A. Top Unigrams**

A table containing:

ngram | tfidf_score

showing the highest-scoring single-word terms.

**B. Top Bigrams**

Important two-word phrases were identified, such as potential skill phrases like:

machine learning

data analysis

social media

business development

**C. Top Trigrams**

Important three-word phrases were identified, including potential technical skill phrases such as:

natural language processing

**D. Multi-Word Skill Candidates**

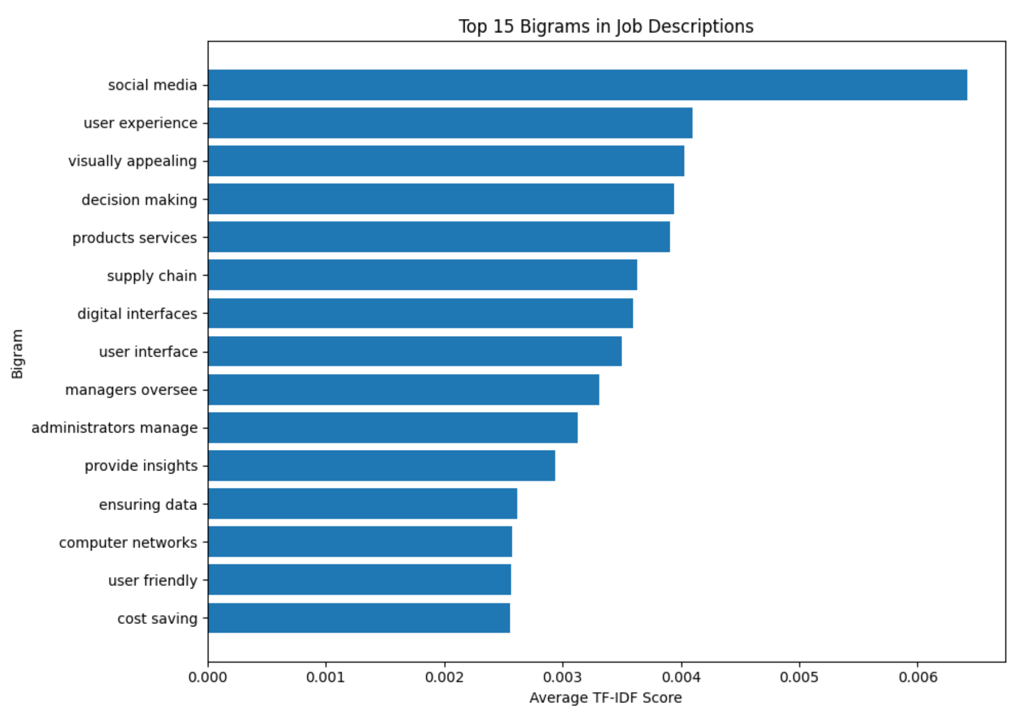
Bigrams and trigrams were compared with the existing taxonomy to identify **potential missing multi-word skills**.

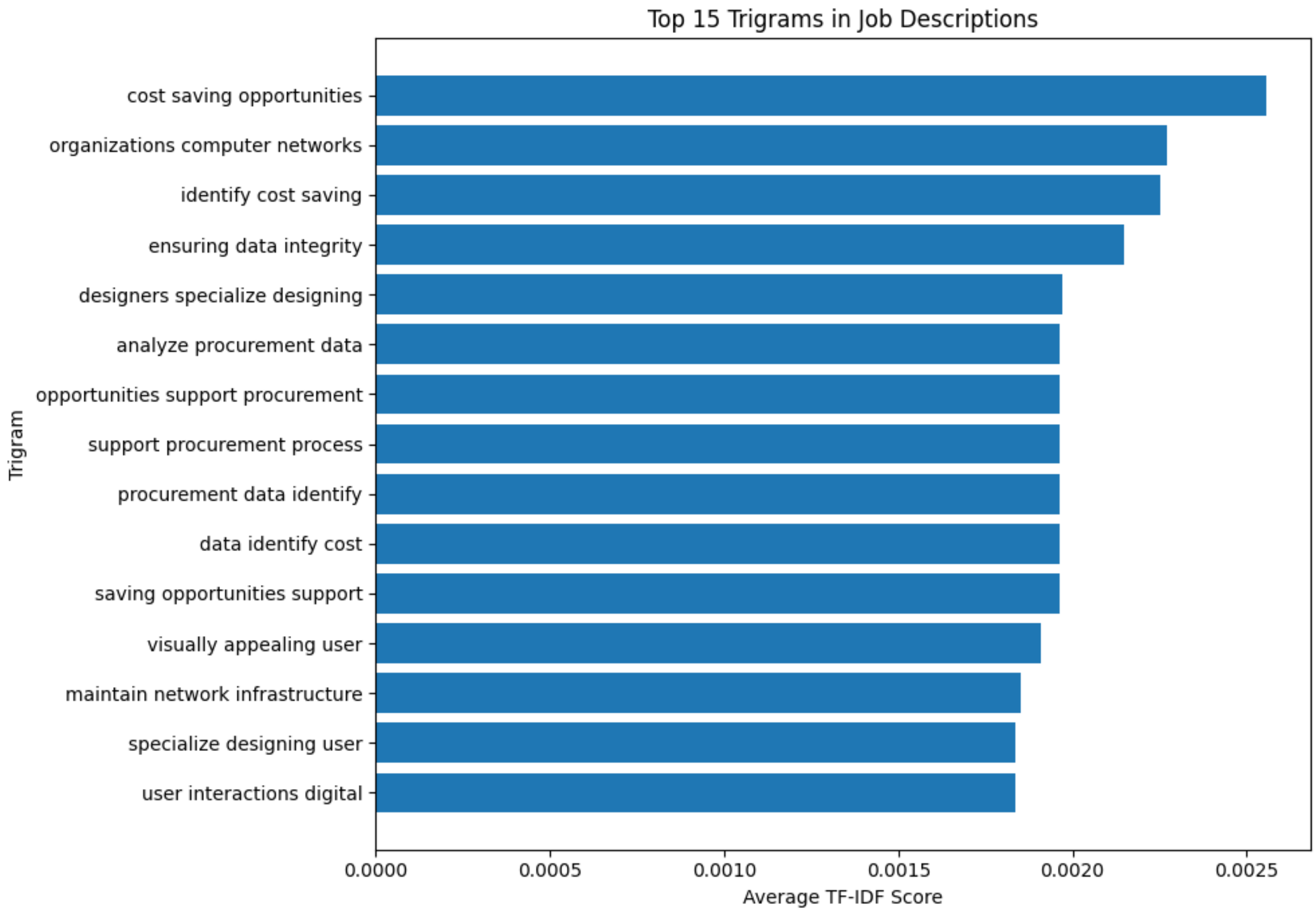
## 7. Visualization

Bar charts were created for:

- Top 15 Bigrams
- Top 15 Trigrams

These visualizations show which multi-word terms received the highest average TF-IDF scores.






## 8. Deliverables

The following files/results were produced:

NLP_NGram_Analysis.ipynb

ngram_analysis_results.csv

unigrams.csv

bigrams.csv

trigrams.csv

## 9. Outcome

Successfully implemented N-Gram analysis on the 5,000 job descriptions using TF-IDF. Unigrams, bigrams, and trigrams were generated to identify important single-word and multi-word terms. The extracted multi-word terms were compared with the existing skill taxonomy to identify potential missing skill phrases and improve the overall skill extraction system.

## 10. Learning

Through this task, I learned:

- What N-Grams are and why they are useful in NLP.
- Difference between unigrams, bigrams, and trigrams.
- How to generate N-Grams using TfidfVectorizer.
- How TF-IDF can help identify important terms.
- How multi-word phrases can improve skill discovery.
- How N-Gram analysis can support **skill taxonomy improvement**.

# DAY 11 — Word Embeddings and Semantic Similarity

## 1. Objective

The objective of Day 11 was to improve the job-skill extraction system by introducing semantic similarity.

The earlier rule-based/taxonomy approach primarily depended on exact words and predefined synonyms. This can fail when two expressions have the same meaning but different wording.

**For example:**

- ML
- Machine Learning

Both refer to the same concept, but simple keyword matching may not always recognize them as equivalent.

Therefore, Day 11 introduced the concept of word/sentence embeddings and semantic similarity.

## 2. Problem with Exact Matching

The taxonomy-based extractor searches for predefined skills and their synonyms.

For example:

Machine Learning

ML

machine-learning

are explicitly included in the taxonomy.

However, job descriptions may contain variations that are not present in the dictionary.

Examples:

machine learning techniques

ML models

predictive modeling

language processing

A simple dictionary-based approach may fail to identify some semantically related expressions.

## 3. Approach Used

The Day 11 workflow was:

Job Description

↓

Candidate Phrase Generation

↓

Generate Unigrams / Bigrams / Trigrams

↓

Skill Dictionary

↓

Create Embeddings

↓

Calculate Semantic Similarity

↓

Identify Possible Skills

Candidate phrases were generated using CountVectorizer.

Example:

- machine learning
- power bi
- data science
- natural language processing

## 4. Candidate Phrase Generation

The following approach was used to generate candidate phrases:

```
from sklearn.feature_extraction.text import CountVectorizer


def generate_phrases(text):


    if pd.isna(text):
        return []


    text = str(text).lower()
```

```
    vectorizer = CountVectorizer(
        ngram_range=(1, 3),
        stop_words="english"
    )

    vectorizer.fit([text])

    phrases = vectorizer.get_feature_names_out()

    return list(phrases)
```

The function generates:

- Unigrams
- Bigrams
- Trigrams

from the job description.

## 5. Sample Test

The function was tested on a job description using:

```
phrases = generate_phrases(
    df.iloc[0]["job_description_matching"]
)

print(phrases[:50])
```

**Output**

| | matched_phrase | canonical_skill | category | similarity |
|---|---|---|---|---|
| 0 | analyze | Data Analysis | Data Analytics | 0.6135 |
| 1 | analyze social | Social Work | Social Services | 0.6036 |
| 2 | analyze social media | Social Media | Digital Marketing | 0.6764 |
| 3 | brand | Brand Marketing | Digital Marketing | 0.7262 |
| 4 | brand awareness | Brand Marketing | Digital Marketing | 0.7432 |
| 5 | brand awareness engagement | Brand Marketing | Digital Marketing | 0.6982 |
| 6 | content | Content Writing | Digital Marketing | 0.6786 |
| 7 | drive brand awareness | Brand Marketing | Digital Marketing | 0.6063 |
| 8 | managers | Leadership | Soft Skills | 0.6387 |
| 9 | media | Social Media | Digital Marketing | 0.6092 |
| 10 | organizations social | Social Work | Social Services | 0.6951 |
| 11 | organizations social media | Social Media | Digital Marketing | 0.7325 |
| 12 | social | Social Work | Social Services | 0.7093 |
| 13 | social media | Social Media | Digital Marketing | 1.0000 |
| 14 | social media managers | Social Media | Digital Marketing | 0.6261 |
| 15 | social media metrics | Social Media | Digital Marketing | 0.6114 |
| 16 | social media presence | Social Media | Digital Marketing | 0.7073 |

For example, the output contains candidate words/phrases extracted from the job description.

## 6. Semantic Similarity Concept

The next step of Day 11 was to compare the meaning of job-description phrases with skills from the taxonomy.

Conceptually:

"ML"

↓

Embedding

↓

Vector

↓

Cosine Similarity

↓

"Machine Learning"

↓

High similarity

The higher the similarity score, the more semantically related the two phrases are.

## 7. Expected Semantic Similarity Example

For documentation, you can show the concept as:

| **Phrase 1** | **Phrase 2** | **Expected Relationship** |
|---|---|---|
| ML | Machine Learning | High |
| NLP | Natural Language Processing | High |
| Power BI | Business Intelligence | Related |
| AWS | Amazon Web Services | High |
| PyTorch | PyTorch | Very High |

## 8. Day 11 Outcome

The Day 11 experiment demonstrated that semantic approaches can help overcome some limitations of exact dictionary matching.

The major learning was:

Exact matching looks for the presence of predefined terms, whereas embeddings attempt to capture semantic relationships between words and phrases.

This provides a foundation for improving the skill extractor beyond simple keyword matching.

## 9. Day 11 Key Learnings

- Understood the limitations of exact skill matching.
- Learned the concept of embeddings.
- Learned how candidate phrases can be generated using n-grams.

- Understood semantic similarity.
- Learned how cosine similarity can be used to compare embeddings.
- Understood how semantic matching can potentially identify variations of technical skills.

| | job_title | additional_semantic_skills |
|---|---|---|
| 0 | Digital Marketing Specialist | {Social Work, Leadership, Content Writing, Data Analysis} |
| 1 | Web Developer | {UI Design, Backend Development, Engineering Design, Software Development, UX Design, Visual Design} |
| 2 | Operations Manager | {Process Improvement, Control Systems, Business Development, Customer Service, Leadership, Quality Assurance, Legal Compliance} |
| 3 | Network Engineer | {Engineering Design, Communication, Problem Solving, Cloud Security, Software Troubleshooting, Computer Networking, UX Design, Electrical Engineering, Network Engineering, Network Troubleshooting} |
| 4 | Event Manager | {Event Management, Vendor Management, Operations Management, Event Planning, Conference Management, Budget Management} |
| ... | ... | ... |
| 95 | Teacher | {Student Support, Instructional Design, Training, Classroom Management, Student Assessment} |
| 96 | Inventory Analyst | {Demand Planning, Control Systems, Performance Management, Business Development} |
| 97 | Legal Counsel | {Litigation, Legal Administration, Customer Acquisition, Family Law, Legal Support, Business Development, Counseling} |
| 98 | Database Developer | {LangGraph, Engineering Design, SQL Server, Big Data, Performance Testing, Database Management, Software Development, Data Structures, UX Design, Content Writing, Network Security} |
| 99 | Procurement Specialist | {Sales, Process Improvement, Operations Management, Customer Service, Leadership, Negotiation} |

100 rows × 2 columns

# DAY 12 — Named Entity Recognition

## 1. Objective

The objective of Day 12 was to understand **Named Entity Recognition (NER)** and investigate whether a standard NLP NER model can identify technical skills from job descriptions.

The experiment was then extended using **custom entities** designed specifically for the job-skill extraction project.

## 2. Standard NER

Named Entity Recognition is an NLP technique used to identify important entities in text.

Typical entities include:

- PERSON
- ORG
- GPE
- DATE
- LOCATION

However, technical skills do not always belong to these standard categories.

For example:

- Python
- SQL
- AWS
- PostgreSQL
- TensorFlow
- Power BI

are important entities for a job-description analysis system, but a general-purpose NER model may not recognize them as technical skills.

## 3. Why Standard NER Can Fail

The experiment demonstrated an important real-world NLP problem.

General-purpose NER models are usually trained to identify common entities such as:

People

Organizations

Locations

Dates

Countries

They are not specifically trained to identify:

- Programming languages
- Databases
- Cloud platforms
- BI tools
- Machine-learning frameworks

Therefore, domain-specific/custom NER is required for a technical skill extraction system.

## 4. Custom Entity Categories

Custom entity categories were created based on the project's skill taxonomy.

| Skill/Technology | Custom Entity |
|---|---|
| Python | SKILL |
| SQL | DATABASE |
| PostgreSQL | DATABASE |
| AWS | CLOUD_PLATFORM |
| TensorFlow | TECHNOLOGY |
| Power BI | TOOL |

The exact mapping depends on the custom taxonomy used in the project.

## 5. Custom NER Using EntityRuler

A spaCy EntityRuler was used to create custom patterns from the existing skill taxonomy.

The workflow was:

Skill Taxonomy

↓

Synonyms / Aliases

↓

Custom NER Patterns

↓

spaCy EntityRuler

↓

Job Description

↓

Detected Entities

## 6. Initial Custom NER Output

The initial experiment produced output such as:

We → CLOUD_PLATFORM

looking for a Data Analyst → SKILL

Python, SQL, → TECHNOLOGY

PostgreSQL and AWS experience → SKILL

Machine → TECHNOLOGY

TensorFlow → SKILL

**Interpretation**

This output was useful because it showed that the NER system was detecting text spans, but some spans were assigned incorrect entity labels.

For example:

We → CLOUD_PLATFORM

is clearly incorrect.

Similarly:

PostgreSQL and AWS experience → SKILL

contains multiple concepts that should ideally be separated.

## 7. Improved Custom NER

The custom EntityRuler was then constructed using the skill taxonomy and category-to-entity mapping.

The objective was to produce results closer to:

Python → SKILL

SQL → DATABASE

PostgreSQL → DATABASE

AWS → CLOUD_PLATFORM

TensorFlow → TECHNOLOGY

Power BI → TOOL

while avoiding incorrect entities such as:

We → CLOUD_PLATFORM

```
('python', {'entities': [(0, 6, 'TECHNOLOGY')]})
('python programming', {'entities': [(0, 18, 'TECHNOLOGY')]})
('python development', {'entities': [(0, 18, 'TECHNOLOGY')]})
('python scripting', {'entities': [(0, 16, 'TECHNOLOGY')]})
('python language', {'entities': [(0, 15, 'TECHNOLOGY')]})
('java', {'entities': [(0, 4, 'TECHNOLOGY')]})
('java programming', {'entities': [(0, 16, 'TECHNOLOGY')]})
('java development', {'entities': [(0, 16, 'TECHNOLOGY')]})
('java language', {'entities': [(0, 13, 'TECHNOLOGY')]})
('c', {'entities': [(0, 1, 'TECHNOLOGY')]})
('c programming', {'entities': [(0, 13, 'TECHNOLOGY')]})
('c language', {'entities': [(0, 10, 'TECHNOLOGY')]})
('c++', {'entities': [(0, 3, 'TECHNOLOGY')]})
('cpp', {'entities': [(0, 3, 'TECHNOLOGY')]})
('c plus plus', {'entities': [(0, 11, 'TECHNOLOGY')]})
('c++ programming', {'entities': [(0, 15, 'TECHNOLOGY')]})
('c#', {'entities': [(0, 2, 'TECHNOLOGY')]})
('c sharp', {'entities': [(0, 7, 'TECHNOLOGY')]})
('csharp', {'entities': [(0, 6, 'TECHNOLOGY')]})
('c# programming', {'entities': [(0, 14, 'TECHNOLOGY')]})
('javascript', {'entities': [(0, 10, 'TECHNOLOGY')]})
('java script', {'entities': [(0, 11, 'TECHNOLOGY')]})
('js', {'entities': [(0, 2, 'TECHNOLOGY')]})
('javascript programming', {'entities': [(0, 22, 'TECHNOLOGY')]})
('typescript', {'entities': [(0, 10, 'TECHNOLOGY')]})
('type script', {'entities': [(0, 11, 'TECHNOLOGY')]})
('ts', {'entities': [(0, 2, 'TECHNOLOGY')]})
('go', {'entities': [(0, 2, 'TECHNOLOGY')]})
('golang', {'entities': [(0, 6, 'TECHNOLOGY')]})
('go programming', {'entities': [(0, 14, 'TECHNOLOGY')]})
```

## 8. Custom NER Applied to Dataset

The custom NER pipeline was applied to the job-description column:

```
df["custom_ner_entities"] = (
    df["job_description_matching"]
    .apply(extract_custom_entities)
)
```

The resulting column stores the entities detected from each job description.

Example structure:

```
[
  {
    "entity": "Python",
    "label": "SKILL"
  },
  {
    "entity": "AWS",
    "label": "CLOUD_PLATFORM"
  }
]
```

## 9. Readable NER Output

The entities were converted into a readable format:

Python → SKILL | SQL → DATABASE | AWS → CLOUD_PLATFORM

**Output**

```
('Strong skills in python are preferred.', {'entities': [(17, 23, 'TECHNOLOGY')]})
('Experience with python programming is required.', {'entities': [(16, 34, 'TECHNOLOGY')]})
('Candidates should have knowledge of python development.', {'entities': [(36, 54, 'TECHNOLOG
('Experience working with java is desirable.', {'entities': [(24, 28, 'TECHNOLOGY')]})
('Strong skills in java programming are preferred.', {'entities': [(17, 33, 'TECHNOLOGY')]})
('Strong skills in java development are preferred.', {'entities': [(17, 33, 'TECHNOLOGY')]})
('The role requires experience using c.', {'entities': [(26, 27, 'TECHNOLOGY')]})
('Knowledge of c programming is an advantage.', {'entities': [(13, 26, 'TECHNOLOGY')]})
('Experience working with c language is desirable.', {'entities': [(24, 34, 'TECHNOLOGY')]})
('Candidates should have knowledge of c++.', {'entities': [(36, 39, 'TECHNOLOGY')]})
('Strong skills in cpp are preferred.', {'entities': [(17, 20, 'TECHNOLOGY')]})
('Knowledge of c plus plus is an advantage.', {'entities': [(13, 24, 'TECHNOLOGY')]})
('Knowledge of c# is an advantage.', {'entities': [(13, 15, 'TECHNOLOGY')]})
('Knowledge of c sharp is an advantage.', {'entities': [(13, 20, 'TECHNOLOGY')]})
('Experience with csharp is required.', {'entities': [(16, 22, 'TECHNOLOGY')]})
('The role requires experience using javascript.', {'entities': [(35, 45, 'TECHNOLOGY')]})
('Experience working with java script is desirable.', {'entities': [(24, 35, 'TECHNOLOGY')]})
('Strong skills in js are preferred.', {'entities': [(17, 19, 'TECHNOLOGY')]})
('Experience working with typescript is desirable.', {'entities': [(24, 34, 'TECHNOLOGY')]})
('Experience with type script is required.', {'entities': [(16, 27, 'TECHNOLOGY')]})
```

This is one of the **best outputs to include in the Day 12 report** because it clearly demonstrates the working of custom NER.

| job_title | custom_ner_readable |
|---|---|
| Digital Marketing Specialist | social media → SKILL \| social media → SKILL \| social media → SKILL \| brand awareness → SKILL |
| Web Developer | web developers → TECHNOLOGY |
| Operations Manager | quality control → SKILL \| quality control → SKILL |
| Network Engineer | wireless network → SKILL \| wireless network → SKILL \| wireless connectivity → SKILL |
| Event Manager | logistics → SKILL \| budgeting → SKILL |
| Software Tester | quality assurance → SKILL |
| Teacher | teacher → SKILL \| lesson plans → SKILL \| instruction → SKILL |
| UX/UI Designer | user interface designers → TOOL \| digital interfaces → TOOL |
| UX/UI Designer | interaction designers → TOOL \| user interactions → TOOL \| digital interfaces → TOOL \| user experiences → TOOL |
| Wedding Planner | budget management → SKILL |
| QA Analyst | performance testing → SKILL |
| Litigation Attorney | family law → SKILL |
| Mechanical Engineer | mechanical design → SKILL \| mechanical systems → SKILL \| manufacturing → SKILL |
| Network Administrator | computer networks → SKILL |
| Account Manager | sales → SKILL \| account manager → SKILL \| sales → SKILL |
| Brand Manager | product brand → SKILL \| brand identity → SKILL |
| Social Worker | social and emotional well-being → SKILL \| connect families with resources → SKILL |
| Social Media Coordinator | social media → SKILL |
| Email Marketing Specialist | email deliverability → SKILL \| email list → SKILL |
| HR Generalist | recruitment → SKILL \| onboarding → SKILL \| training → SKILL |

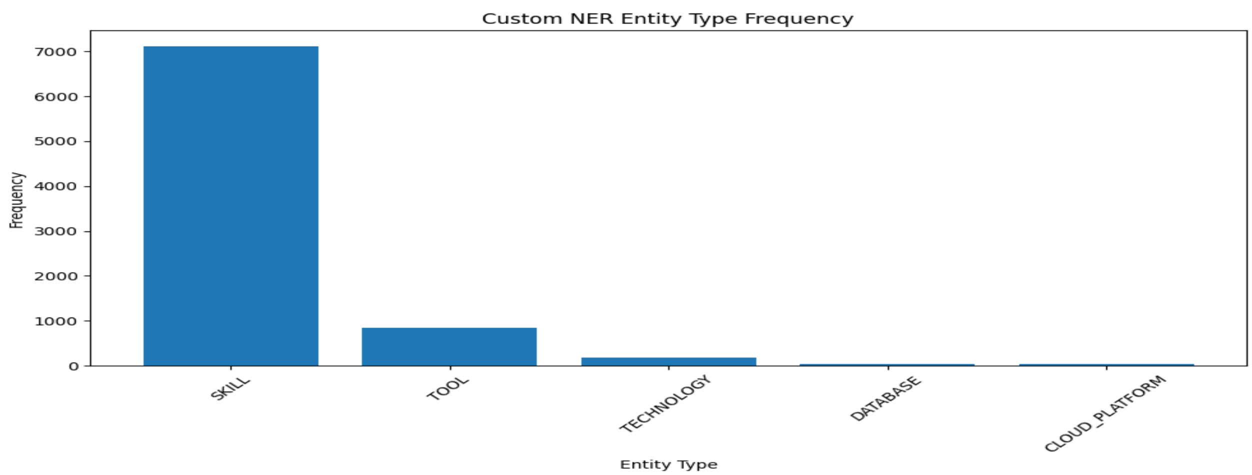
## 10. Entity Frequency Analysis

Entity types were counted to understand which types appeared most frequently in the dataset.

The output was generated using:

display(entity_frequency)

**Output**




## 11. Jobs Without Custom NER Entities

The project also checked job descriptions where no custom entities were detected.

The code calculated:

```
print(
   "Jobs with no custom NER entities:",
   len(no_entity_jobs)
)
```

and:

```
print(
   "Percentage:",
   round(
      len(no_entity_jobs) / len(df) * 100,
      2
   ),
   "%"
)
```

```
Jobs with no custom NER entities: 1165
Percentage: 23.3 %
```

| | job_title | job_description_matching |
|---|---|---|
| 55 | Key Account Manager | strategic account managers nurture relationships with key clients. they develop account strategies, address client needs, and seek opportunities for business growth and expansion. |
| 75 | Technical Writer | documentation specialists create and manage documents, manuals, or technical materials. they ensure accuracy, consistency, and accessibility of documentation. |
| 78 | Key Account Manager | strategic account managers nurture relationships with key clients. they develop account strategies, address client needs, and seek opportunities for business growth and expansion. |
| 84 | Environmental Consultant | sustainability consultants advise organizations on sustainable practices and strategies. they assess environmental impact, recommend sustainability initiatives, and assist in the development and i... |
| 89 | Nurse Practitioner | acute care nurse practitioners focus on managing acute and critical medical conditions, often working in hospitals or emergency care settings. |

Percentage: 23%

This is important because it helps evaluate the coverage of the custom NER approach.

## 12. Comparison With Previous Skill Extraction

The Day 12 experiment can also be compared with the previous taxonomy-based extractor.

```
display(
   df[
      [
         "job_title",
         "extracted_skills",
         "custom_ner_entities"
      ]
   ].head(20)
)
```

**Output**

| | | | |
|---|---|---|---|
| 1 | Web Developer | [{'category': 'Web Development', 'canonical_skill': 'Web Development', 'matched_synonym': 'web developers'}] | [{'entity': 'web developers', 'label': 'TECHNOLOGY'}] |
| 2 | Operations Manager | [{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality control'}, {'category': 'Operations', 'canonical_skill': 'Quality Control', 'matched_synony... | [{'entity': 'quality control', 'label': 'SKILL'}, {'entity': 'quality control', 'label': 'SKILL'}] |
| 3 | Network Engineer | [{'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym': 'wireless network'}] | [{'entity': 'wireless network', 'label': 'SKILL'}, {'entity': 'wireless network', 'label': 'SKILL'}, {'entity': 'wireless connectivity', 'label': 'SKILL'}] |
| 4 | Event Manager | [{'category': 'Event Management', 'canonical_skill': 'Logistics', 'matched_synonym': 'logistics'}, {'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_synonym': 'budgetin... | [{'entity': 'logistics', 'label': 'SKILL'}, {'entity': 'budgeting', 'label': 'SKILL'}] |
| 5 | Software Tester | [{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Assurance', 'matched_synonym': 'quality assurance'}, {'category': 'Operations', 'canonical_skill': 'Quality Control', 'matched_sy... | [{'entity': 'quality assurance', 'label': 'SKILL'}] |
| 6 | Teacher | [{'category': 'Education & Teaching', 'canonical_skill': 'Teaching', 'matched_synonym': 'teacher'}, {'category': 'Education & Teaching', 'canonical_skill': 'Lesson Planning', 'matched_synonym': 'l... | [{'entity': 'teacher', 'label': 'SKILL'}, {'entity': 'lesson plans', 'label': 'SKILL'}, {'entity': 'instruction', 'label': 'SKILL'}] |
| 7 | UX/UI Designer | [{'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'user interface designers'}] | [{'entity': 'user interface designers', 'label': 'TOOL'}, {'entity': 'digital interfaces', 'label': 'TOOL'}] |
| 8 | UX/UI Designer | [{'category': 'UI/UX Design', 'canonical_skill': 'UX Design', 'matched_synonym': 'interaction designers'}, {'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'digital ... | [{'entity': 'interaction designers', 'label': 'TOOL'}, {'entity': 'user interactions', 'label': 'TOOL'}, {'entity': 'digital interfaces', 'label': 'TOOL'}, {'entity': 'user experiences', 'label': ... |
| 9 | Wedding Planner | [{'category': 'Event Management', 'canonical_skill': 'Budget Management', 'matched_synonym': 'budget management'}, {'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_syn... | [{'entity': 'budget management', 'label': 'SKILL'}] |
| 10 | QA Analyst | [{'category': 'Software Testing & QA', 'canonical_skill': 'Performance Testing', 'matched_synonym': 'performance testing'}] | [{'entity': 'performance testing', 'label': 'SKILL'}] |
| 11 | Litigation Attorney | [{'category': 'Legal', 'canonical_skill': 'Family Law', 'matched_synonym': 'family law'}] | [{'entity': 'family law', 'label': 'SKILL'}] |

This demonstrates how the project is evolving:

Dictionary Matching

↓

Regex / Phrase Matching

↓

N-Gram Analysis

↓

Semantic Similarity

↓

Custom NER

## 13. Day 12 Outcome

The Day 12 experiment demonstrated that standard NER is not sufficient for technical skill extraction because general NER models are not specifically trained on technology-related entities.

A custom NER approach was therefore implemented using spaCy's EntityRuler and the project's skill taxonomy.

The experiment also showed that **entity detection and correct entity classification are two different problems**. Initial results contained incorrect labels, demonstrating the importance of carefully designed domain-specific patterns.

## 14. Day 12 Key Learnings

- Understood Named Entity Recognition.
- Tested standard NER on job descriptions.
- Identified limitations of general-purpose NER.
- Created domain-specific entity categories.
- Used the existing skill taxonomy to generate custom patterns.
- Implemented custom NER using spaCy EntityRuler.
- Applied custom NER to job descriptions.
- Analyzed entity frequencies.
- Identified job descriptions with no detected entities.
- Compared custom NER results with previous skill extraction.

# DAY 13 — ML Skill Classifier

## Objective

Build a supervised ML model to classify job descriptions into skill categories using **TF-IDF + Logistic Regression**.

### Steps Performed

1. Prepared labeled job-description data from the existing skill taxonomy.
2. Removed records without skill categories.
3. Split data into **80% training / 20% testing**.
4. Converted text into numerical features using **TF-IDF** with unigrams, bigrams, and trigrams.
5. Trained a **Logistic Regression** classifier.
6. Predicted skill categories for test and new job descriptions.
7. Evaluated the model using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
    - Confusion Matrix
8. Checked prediction confidence.
9. Saved the trained model.

## Key Output

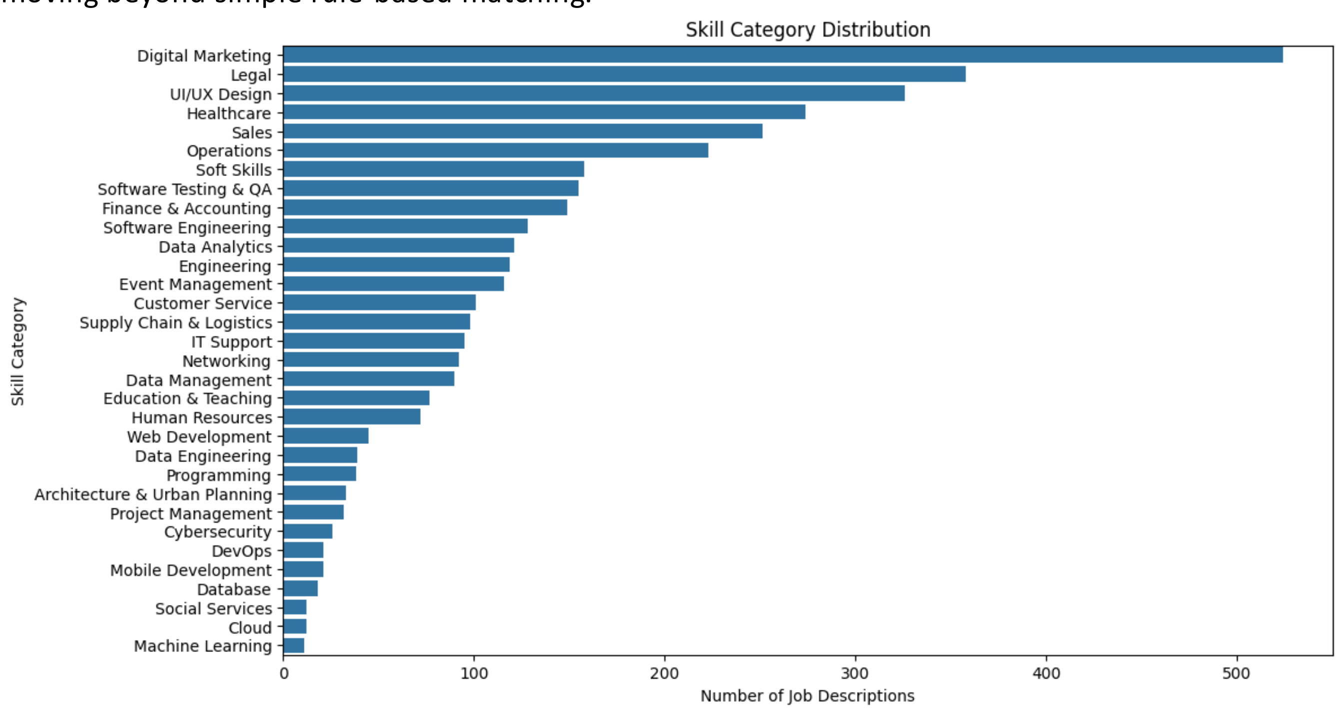
TF-IDF → Logistic Regression → Skill Category

## Model file:

skill_classifier.pkl

## Outcome

Successfully added a **Machine Learning-based classification layer** to the job-skill extraction pipeline, moving beyond simple rule-based matching.



```
print("Original records:", len(df))
print("ML records:", len(ml_df))
```

```
Original records: 5000
ML records: 3835
```

Figure: Original Record vs ML Records

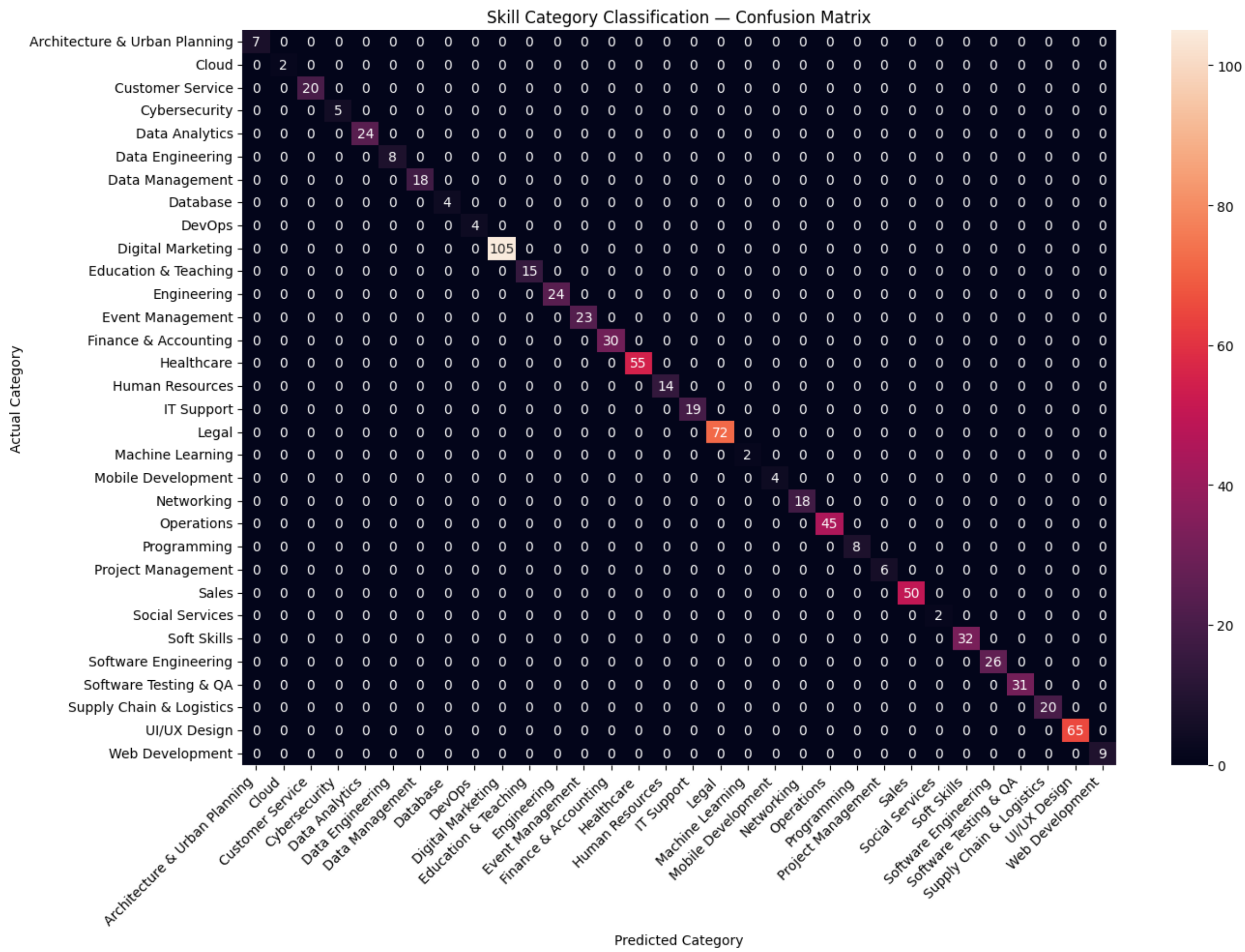
```
#Training Logistic Regression
classifier = LogisticRegression(
    max_iter=2000,
    random_state=42
)

classifier.fit(
    X_train_tfidf,
    y_train
)
```

```
LogisticRegression
LogisticRegression(max_iter=2000, random_state=42)
```

Figure: Logistic Regression Training

```
#Accuracy
accuracy = accuracy_score(
    y_test,
    y_pred
)

print(
    f"Model Accuracy: {accuracy:.2%}"
)
```

```
Model Accuracy: 100.00%
```



Skill Category Classification — Confusion Matrix
Actual Category
Predicted Category
Architecture & Urban Planning
Cloud
Customer Service
Cybersecurity
Data Analytics
Data Engineering
Data Management
Database
DevOps
Digital Marketing
Education & Teaching
Engineering
Event Management
Finance & Accounting
Healthcare
Human Resources
IT Support
Legal
Machine Learning
Mobile Development
Networking
Operations
Programming
Project Management
Sales
Social Services
Soft Skills
Software Engineering
Software Testing & QA
Supply Chain & Logistics
UI/UX Design
Web Development
7 2 20 5 24 8 18 4 4 105 15 24 23 30 55 14 19 72 2 4 18 45 8 6 50 2 32 26 31 20 65 9
0
20
40
60
80
100

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Architecture & Urban Planning | 1.00 | 1.00 | 1.00 | 7 |
| Cloud | 1.00 | 1.00 | 1.00 | 2 |
| Customer Service | 1.00 | 1.00 | 1.00 | 20 |
| Cybersecurity | 1.00 | 1.00 | 1.00 | 5 |
| Data Analytics | 1.00 | 1.00 | 1.00 | 24 |
| Data Engineering | 1.00 | 1.00 | 1.00 | 8 |
| Data Management | 1.00 | 1.00 | 1.00 | 18 |
| Database | 1.00 | 1.00 | 1.00 | 4 |
| DevOps | 1.00 | 1.00 | 1.00 | 4 |
| Digital Marketing | 1.00 | 1.00 | 1.00 | 105 |
| Education & Teaching | 1.00 | 1.00 | 1.00 | 15 |
| Engineering | 1.00 | 1.00 | 1.00 | 24 |
| Event Management | 1.00 | 1.00 | 1.00 | 23 |
| Finance & Accounting | 1.00 | 1.00 | 1.00 | 30 |
| Healthcare | 1.00 | 1.00 | 1.00 | 55 |
| Human Resources | 1.00 | 1.00 | 1.00 | 14 |
| IT Support | 1.00 | 1.00 | 1.00 | 19 |
| Legal | 1.00 | 1.00 | 1.00 | 72 |
| Machine Learning | 1.00 | 1.00 | 1.00 | 2 |
| Mobile Development | 1.00 | 1.00 | 1.00 | 4 |
| Networking | 1.00 | 1.00 | 1.00 | 18 |
| Operations | 1.00 | 1.00 | 1.00 | 45 |
| Programming | 1.00 | 1.00 | 1.00 | 8 |
| Project Management | 1.00 | 1.00 | 1.00 | 6 |
| Sales | 1.00 | 1.00 | 1.00 | 50 |
| Social Services | 1.00 | 1.00 | 1.00 | 2 |
| Soft Skills | 1.00 | 1.00 | 1.00 | 32 |
| Software Engineering | 1.00 | 1.00 | 1.00 | 26 |
| Software Testing & QA | 1.00 | 1.00 | 1.00 | 31 |
| Supply Chain & Logistics | 1.00 | 1.00 | 1.00 | 20 |
| UI/UX Design | 1.00 | 1.00 | 1.00 | 65 |
| Web Development | 1.00 | 1.00 | 1.00 | 9 |
| accuracy | | | 1.00 | 767 |
| macro avg | 1.00 | 1.00 | 1.00 | 767 |
| weighted avg | 1.00 | 1.00 | 1.00 | 767 |

```
JOB DESCRIPTION:
We need a data analyst with Power BI, Excel and data visualization skills.

PREDICTED CATEGORY:
Data Analytics
================================================================================
```

| | job_description | actual_category | predicted_category |
|---|---|---|---|
| 0 | budget analysts manage and analyze financial data to develop, monitor, and report on budgets. they work with departments to ensure budget compliance, identify cost-saving opportunities, and provid... | Legal | Legal |
| 1 | frontend web designers create the visual elements and user interfaces of websites. they use html, css, and javascript to design responsive, user-friendly web pages, ensuring a seamless and visuall... | Web Development | Web Development |
| 2 | administrative managers oversee office operations. they manage administrative staff, coordinate tasks, and ensure smooth office functioning to support the organizations goals. | Operations | Operations |
| 3 | oversee sales operations within a specific region, develop sales plans, and manage regional sales teams. | Sales | Sales |
| 4 | wealth advisors provide financial advice to clients. they assess financial goals, create investment strategies, and offer guidance on wealth management and financial planning. | Finance & Accounting | Finance & Accounting |
| 5 | seo specialists optimize websites for search engines to improve online visibility. they conduct keyword research, optimize content, and implement seo strategies to increase organic traffic and ran... | Digital Marketing | Digital Marketing |
| 6 | a key account executive manages strategic accounts for a company. they build strong client relationships and work to expand business opportunities within these key accounts. | Sales | Sales |
| 7 | user interface designers focus on the visual and interactive aspects of digital interfaces. they design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interf... | UI/UX Design | UI/UX Design |
| 8 | technical seo specialists optimize website infrastructure, addressing issues like site speed, mobile-friendliness, and structured data to enhance search engine visibility. | Digital Marketing | Digital Marketing |
| 9 | desktop support technicians troubleshoot and maintain desktop computer systems, resolving hardware and software problems for users. | IT Support | IT Support |
| 10 | procurement managers oversee procurement processes, managing vendor relationships, negotiating contracts, and ensuring cost-effective purchasing of goods and services. | Operations | Operations |
| 11 | an administrative assistant provides administrative support to the organization, including scheduling, document management, and assisting in day-to-day operations. | Operations | Operations |
| 12 | coordinate corporate events, conferences, and meetings, handle logistics, and ensure a seamless experience for attendees. | Event Management | Event Management |
| 13 | monitor and improve email deliverability rates, troubleshoot delivery issues, and maintain a clean email list. | Digital Marketing | Digital Marketing |
| 14 | corporate attorneys provide legal counsel and representation to organizations. they handle legal matters, contracts, mergers, acquisitions, and ensure compliance with laws and regulations relevant... | Legal | Legal |

Figure: Actual Category vs predicted category

## DAY 13 Summary

```
    print("\nSaved File:")
    print("skill_classifier.pkl")
```

```
============================================================
DAY 13 — MODEL SUMMARY
============================================================
Dataset Size: 3835
Training Size: 3068
Testing Size: 767
Number of Categories: 32
Accuracy: 100.00%

Model:
TF-IDF + Logistic Regression

Saved File:
skill_classifier.pkl
```

DAY 14 - Skill Extraction Evaluation & Error Analysis

## Objective:

Evaluate the Day 13 job-title classification model on unseen test data and identify classification errors.

## Tasks completed:

- Test-set evaluation
- Accuracy calculation
- Precision, recall and F1-score
- Confusion matrix
- Error analysis
- Incorrect prediction analysis
- Class distribution analysis
- New job-description testing
- Prediction confidence analysis
- Top-3 prediction analysis

```
accuracy = accuracy_score(y_test, y_pred)

print(f"Test Accuracy: {accuracy:.2%}")
```

```
Test Accuracy: 100.00%
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Architecture & Urban Planning | 1.00 | 1.00 | 1.00 | 7 |
| Cloud | 1.00 | 1.00 | 1.00 | 2 |
| Customer Service | 1.00 | 1.00 | 1.00 | 20 |
| Cybersecurity | 1.00 | 1.00 | 1.00 | 5 |
| Data Analytics | 1.00 | 1.00 | 1.00 | 24 |
| Data Engineering | 1.00 | 1.00 | 1.00 | 8 |
| Data Management | 1.00 | 1.00 | 1.00 | 18 |
| Database | 1.00 | 1.00 | 1.00 | 4 |
| DevOps | 1.00 | 1.00 | 1.00 | 4 |
| Digital Marketing | 1.00 | 1.00 | 1.00 | 105 |
| Education & Teaching | 1.00 | 1.00 | 1.00 | 15 |

```
DESCRIPTION:

We are looking for a Data Engineer with experience in Python,
SQL, PostgreSQL, AWS, Spark, PySpark, Docker and Kubernetes.


CUSTOM SKILL EXTRACTION:
python → Python → Programming
sql → SQL → Database
postgresql → PostgreSQL → Database
aws → AWS → Cloud
spark → Apache Spark → Data Engineering
pyspark → PySpark → Data Engineering
docker → Docker → DevOps
kubernetes → Kubernetes → DevOps
```

| | job_title | job_description_matching | transformer_entities |
|---|---|---|---|
| 0 | Digital Marketing Specialist | social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagem... | [] |
| 1 | Web Developer | frontend web developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. they collaborate with designers and backend developers to create ... | [] |
| 2 | Operations Manager | quality control managers establish and enforce quality standards within an organization. they develop quality control processes, perform inspections, and implement corrective actions to maintain p... | [] |
| 3 | Network Engineer | wireless network engineers design, implement, and maintain wireless network solutions. they optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communicati... | [] |
| 4 | Event Manager | a conference manager coordinates and manages conferences, meetings, and events. they plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to t... | [] |
| 5 | Software Tester | a quality assurance analyst tests software and products to ensure they meet quality standards. they identify defects, report issues, and work with development teams to resolve problems. | [] |
| 6 | Teacher | a classroom teacher educates students in a specific subject or grade level. they create lesson plans, deliver instruction, assess student progress, and foster a positive learning environment. | [] |
| 7 | UX/UI Designer | user interface designers focus on the visual and interactive aspects of digital interfaces. they design layouts, buttons, and other elements to ensure a cohesive and visually appealing user interf... | [] |
| 8 | UX/UI Designer | interaction designers specialize in designing user interactions within digital interfaces. they create meaningful and engaging user experiences by considering user behaviors and system responses. | [] |
| 9 | Wedding Planner | a wedding consultant assists couples in planning and coordinating their weddings, helping them with vendor selection, budget management, and ensuring a seamless wedding day experience. | [] |

| | job_title | taxonomy_skills |
|---|---|---|
| 0 | Digital Marketing Specialist | [{'category': 'Digital Marketing', 'canonical_skill': 'Social Media', 'matched_synonym': 'social media'}, {'category': 'Digital Marketing', 'canonical_skill': 'Brand Marketing', 'matched_synonym':... |
| 1 | Web Developer | [{'category': 'Web Development', 'canonical_skill': 'Web Development', 'matched_synonym': 'web developers'}] |
| 2 | Operations Manager | [{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality control'}] |
| 3 | Network Engineer | [{'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym': 'wireless network'}, {'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym'... |
| 4 | Event Manager | [{'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_synonym': 'budgeting'}, {'category': 'Event Management', 'canonical_skill': 'Logistics', 'matched_synonym': 'logistic... |
| 5 | Software Tester | [{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Assurance', 'matched_synonym': 'quality assurance'}] |
| 6 | Teacher | [{'category': 'Education & Teaching', 'canonical_skill': 'Teaching', 'matched_synonym': 'teacher'}, {'category': 'Education & Teaching', 'canonical_skill': 'Teaching', 'matched_synonym': 'instruct... |
| 7 | UX/UI Designer | [{'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'user interface designers'}, {'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'inter... |
| 8 | UX/UI Designer | [{'category': 'UI/UX Design', 'canonical_skill': 'UI Design', 'matched_synonym': 'digital interfaces'}, {'category': 'UI/UX Design', 'canonical_skill': 'UX Design', 'matched_synonym': 'user experi... |
| 9 | Wedding Planner | [{'category': 'Event Management', 'canonical_skill': 'Budget Management', 'matched_synonym': 'budget management'}] |

```
NLP SKILL EXTRACTION PIPELINE

Traditional NLP
      ↓
Text Preprocessing
      ↓
TF-IDF
      ↓
N-Gram Analysis
      ↓
Word Embeddings
      ↓
NER
      ↓
Custom Skill Taxonomy
      ↓
Transformer NER
      ↓
Transformer vs Custom Skill Extraction
```

## Key learning

Day 14 demonstrated that overall accuracy alone is not sufficient for evaluating an NLP classification model. Error analysis, confusion matrices, class distribution, and prediction confidence provide deeper insight into model performance and help identify cases where job descriptions are semantically similar or insufficiently distinctive.

## DAY 15 — Model Evaluation

### Objective

Evaluate the **custom taxonomy-based skill extraction system** against manually labelled skills and measure its performance.

### Approach

The 100 manually labelled job descriptions were compared with the skills extracted by the custom taxonomy. The following metrics were calculated:

- **True Positive (TP):** Correctly extracted skills
- **False Positive (FP):** Incorrectly extracted skills
- **False Negative (FN):** Actual skills that were missed
- **Precision:** Measures how many predicted skills were correct.
- **Recall:** Measures how many actual skills were successfully extracted.
- **F1 Score:** Harmonic mean of Precision and Recall.
- **Exact-Match Accuracy:** Percentage of job descriptions where the complete actual and predicted skill sets matched exactly.
- **Confusion Matrix:** Skill-level TP, FP, FN and TN analysis.

## Actual Results

| **Metric** | **Result** |
|---|---|
| Total Records | 110 |
| True Positives | 33 |
| False Positives | 143 |
| False Negatives | 85 |
| Precision | **18.75%** |
| Recall | **27.97%** |
| F1 Score | **22.45%** |

| Metric | Result |
|---|---|
| Exact Accuracy | 1.82% |

```
Day 14 shape: (100, 4)

Day 14 columns:
['job_title', 'job_description_matching', 'transformer_entities', 'taxonomy_skills']
```

| | job_title | job_description_matching | transformer_entities | taxonomy_skills |
|---|---|---|---|---|
| 0 | Digital Marketing Specialist | social media managers oversee an organizations social media presence. they create and schedule content, engage with followers, and analyze social media metrics to drive brand awareness and engagem... | [] | [{'category': 'Digital Marketing', 'canonical_skill': 'Social Media', 'matched_synonym': 'social media'}, {'category': 'Digital Marketing', 'canonical_skill': 'Brand Marketing', 'matched_synonym':... |
| 1 | Web Developer | frontend web developers design and implement user interfaces for websites, ensuring they are visually appealing and user-friendly. they collaborate with designers and backend developers to create ... | [] | [{'category': 'Web Development', 'canonical_skill': 'Web Development', 'matched_synonym': 'web developers'}] |
| 2 | Operations Manager | quality control managers establish and enforce quality standards within an organization. they develop quality control processes, perform inspections, and implement corrective actions to maintain p... | [] | [{'category': 'Software Testing & QA', 'canonical_skill': 'Quality Control', 'matched_synonym': 'quality control'}] |
| 3 | Network Engineer | wireless network engineers design, implement, and maintain wireless network solutions. they optimize wireless connectivity, troubleshoot issues, and ensure reliable and secure wireless communicati... | [] | [{'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym': 'wireless network'}, {'category': 'Networking', 'canonical_skill': 'Wireless Networking', 'matched_synonym'... |
| 4 | Event Manager | a conference manager coordinates and manages conferences, meetings, and events. they plan logistics, handle budgeting, liaise with vendors, and ensure the smooth execution of events, catering to t... | [] | [{'category': 'Finance & Accounting', 'canonical_skill': 'Budgeting', 'matched_synonym': 'budgeting'}, {'category': 'Event Management', 'canonical_skill': 'Logistics', 'matched_synonym': 'logistic... |

```
Dataset shape: (100, 3)

Columns:
['job_title', 'job_description_matching', 'actual_skills']
```

<table>
<tr><th></th><th>job_title</th><th>job_description_matching</th><th>actual_skills</th></tr>
<tr><td>0</td><td>Procurement Manager</td><td>analyze procurement data, identify cost-saving opportunities, and support the procurement process.</td><td>procurement|data analysis|cost-saving analysis</td></tr>
<tr><td>1</td><td>Account Executive</td><td>a sales account executive is responsible for acquiring and managing business accounts. they identify sales opportunities, negotiate contracts, and meet revenue targets.</td><td>sales|account management|contract negotiation</td></tr>
<tr><td>2</td><td>Graphic Designer</td><td>ui/ux designers focus on creating user-friendly and visually appealing interfaces for digital products. they conduct user research, design wireframes and prototypes, and collaborate with developme...</td><td>ui/ux design|user research|wireframing|prototyping|collaboration</td></tr>
<tr><td>3</td><td>Paralegal</td><td>assist in real estate transactions, including title searches, closings, and contract preparation.</td><td>real estate transactions|title search|contract preparation</td></tr>
<tr><td>4</td><td>UX/UI Designer</td><td>user experience designers create intuitive and user-friendly digital interfaces. they conduct user research, design prototypes, and work to enhance the overall user experience of websites and appl...</td><td>user research|prototyping|ui/ux design</td></tr>
</table>

```
Actual skill sets created successfully.
```

<table>
<tr><th></th><th>job_title</th><th>actual_skills</th><th>actual_skill_set</th></tr>
<tr><td>0</td><td>Digital Marketing Specialist</td><td>NaN</td><td>{}</td></tr>
<tr><td>1</td><td>Web Developer</td><td>NaN</td><td>{}</td></tr>
<tr><td>2</td><td>Operations Manager</td><td>quality control|process development|inspection</td><td>{process development, inspection, quality control}</td></tr>
<tr><td>3</td><td>Network Engineer</td><td>NaN</td><td>{}</td></tr>
<tr><td>4</td><td>Event Manager</td><td>NaN</td><td>{}</td></tr>
<tr><td>5</td><td>Software Tester</td><td>quality assurance|defect identification|testing</td><td>{testing, defect identification, quality assurance}</td></tr>
<tr><td>6</td><td>Teacher</td><td>NaN</td><td>{}</td></tr>
<tr><td>7</td><td>UX/UI Designer</td><td>NaN</td><td>{}</td></tr>
<tr><td>8</td><td>UX/UI Designer</td><td>interaction design|ux design</td><td>{ux design, interaction design}</td></tr>
<tr><td>9</td><td>Wedding Planner</td><td>NaN</td><td>{}</td></tr>
</table>

```
Predicted skill sets created successfully.
```

| | job_title | actual_skill_set | predicted_skill_set |
|---|---|---|---|
| 0 | Digital Marketing Specialist | {} | {brand marketing, social media} |
| 1 | Web Developer | {} | {web development} |
| 2 | Operations Manager | {process development, inspection, quality control} | {quality control} |
| 3 | Network Engineer | {} | {wireless networking} |
| 4 | Event Manager | {} | {logistics, budgeting} |
| 5 | Software Tester | {testing, defect identification, quality assurance} | {quality assurance} |
| 6 | Teacher | {} | {lesson planning, teaching} |
| 7 | UX/UI Designer | {} | {ui design} |
| 8 | UX/UI Designer | {ux design, interaction design} | {ux design, ui design} |
| 9 | Wedding Planner | {} | {budget management} |

```
True Positives : 33
False Positives: 143
False Negatives: 85
```

## Outcome

The evaluation showed that the current taxonomy has **low precision and recall**, with a relatively high number of false positives and false negatives. This indicates that the skill extraction system requires further improvement in **synonym handling, skill normalization, taxonomy coverage, and entity matching**.

```
======================================
DAY 15 — MODEL EVALUATION
======================================
Total Records     : 110
True Positives    : 33
False Positives   : 143
False Negatives   : 85
Precision         : 0.1875
Recall            : 0.2797
F1 Score          : 0.2245
Exact Accuracy    : 0.0182
```

The results were saved in:

model_comparison.csv

**Day 15 Deliverable:** Model evaluation and performance comparison completed.

# DAY 16 — Error Analysis

## Objective

Identify and analyze the errors made by the skill extraction system after the Day 15 evaluation.

### Approach

The predicted skills were compared with the manually labelled actual skills to identify different types of extraction errors.

## Error Categories

- **False Positive:** Skill predicted but not present in the actual skills.
- **False Negative:** Actual skill present but not extracted.
- **Partial Entity:** Only part of a multi-word skill was extracted.
- **Duplicate Skill:** Same skill extracted multiple times through different synonyms.
- **Synonym Problem:** Different terms representing the same skill were not correctly mapped.
- **Unknown Skill:** A skill was present but missing from the current taxonomy.
- **Spelling Problem:** Skill was missed because of spelling or textual variation.

## Analysis Performed

The project generated:

- Individual FP and FN records
- Most frequent false-positive skills
- Most frequent false-negative skills
- Duplicate skill analysis
- Possible synonym/partial-match cases
- Overall error-category summary

False Positives: 143

| | Job Title | Actual | Predicted |
|---|---|---|---|
| 0 | Digital Marketing Specialist | | brand marketing |
| 1 | Digital Marketing Specialist | | social media |
| 2 | Web Developer | | web development |
| 5 | Network Engineer | | wireless networking |
| 6 | Event Manager | | logistics |
| 7 | Event Manager | | budgeting |
| 10 | Teacher | | lesson planning |
| 11 | Teacher | | teaching |
| 12 | UX/UI Designer | | ui design |
| 13 | UX/UI Designer | | ui design |
| 15 | Wedding Planner | | budget management |

P/FN analysis created successfully.

| | job_title | actual_skill_set | predicted_skill_set | false_positive_skills | false_negative_skills |
|---|---|---|---|---|---|
| 0 | Digital Marketing Specialist | {} | {brand marketing, social media} | {brand marketing, social media} | {} |
| 1 | Web Developer | {} | {web development} | {web development} | {} |
| 2 | Operations Manager | {process development, inspection, quality control} | {quality control} | {} | {process development, inspection} |
| 3 | Network Engineer | {} | {wireless networking} | {wireless networking} | {} |
| 4 | Event Manager | {} | {logistics, budgeting} | {logistics, budgeting} | {} |
| 5 | Software Tester | {testing, defect identification, quality assurance} | {quality assurance} | {} | {testing, defect identification} |
| 6 | Teacher | {} | {lesson planning, teaching} | {lesson planning, teaching} | {} |
| 7 | UX/UI Designer | {} | {ui design} | {ui design} | {} |
| 8 | UX/UI Designer | {ux design, interaction design} | {ux design, ui design} | {ui design} | {interaction design} |
| 9 | Wedding Planner | {} | {budget management} | {budget management} | {} |

```
False Negatives: 85
```

| | Job Title | Actual | Predicted |
|---|---|---|---|
| 3 | Operations Manager | process development | |
| 4 | Operations Manager | inspection | |
| 8 | Software Tester | testing | |
| 9 | Software Tester | defect identification | |
| 14 | UX/UI Designer | interaction design | |
| 22 | Network Administrator | network security | |
| 23 | Network Administrator | network monitoring | |
| 24 | Network Administrator | incident response | |
| 26 | Network Administrator | network security | |
| 27 | Network Administrator | network monitoring | |
| 28 | Network Administrator | incident response | |
| 35 | Social Media Coordinator | content creation | |
| 36 | Social Media Coordinator | social media marketing | |
| 37 | Email Marketing Specialist | email deliverability | |

Rows containing duplicate skills: 37

| | job_title | duplicate_skills |
|---|---|---|
| 3 | Network Engineer | [wireless networking] |
| 6 | Teacher | [teaching] |
| 7 | UX/UI Designer | [ui design] |
| 8 | UX/UI Designer | [ux design] |
| 16 | Brand Manager | [brand marketing] |
| 17 | Social Worker | [emotional well-being] |
| 19 | Email Marketing Specialist | [email marketing] |
| 27 | Purchasing Agent | [supply chain management] |
| 28 | Purchasing Agent | [supply chain management] |
| 30 | Civil Engineer | [structural engineering] |
| 34 | Financial Planner | [taxation] |
| 38 | Electrical Designer | [lighting design] |

| job_title | actual_skill_set | predicted_skill_set | possible_synonym_pairs |
|---|---|---|---|
| UX/UI Designer | {ux design, interaction design} | {ux design, ui design} | [ux design <-> ui design, interaction design <-> ux design, interaction design <-> ui design] |
| Social Media Coordinator | {content creation, social media marketing} | {social media} | [social media marketing <-> social media] |
| Email Marketing Specialist | {email deliverability, troubleshooting, email marketing} | {email marketing} | [email deliverability <-> email marketing] |
| Purchasing Agent | {logistics coordination, supply chain management} | {logistics, operations management, supply chain management} | [logistics coordination <-> logistics, supply chain management <-> operations management] |
| Purchasing Agent | {logistics coordination, supply chain management} | {logistics, operations management, supply chain management} | [logistics coordination <-> logistics, supply chain management <-> operations management] |
| UI Developer | {coding, front-end development, ui design} | {ux design} | [ui design <-> ux design] |
| UI Developer | {coding, front-end development, ui design} | {ux design} | [ui design <-> ux design] |
| Data Analyst | {reporting, dashboarding, data analysis, data visualization} | {data analysis, data visualization} | [data analysis <-> data visualization, data visualization <-> data analysis] |
| Account Executive | {sales, account management, contract negotiation} | {sales, account management, contract law} | [contract negotiation <-> contract law] |
| Account Executive | {sales, account management, contract negotiation} | {sales, account management, contract law} | [contract negotiation <-> contract law] |
| Sales Manager | {sales management, sales planning, team management} | {sales, operations management} | [sales management <-> sales, sales management <-> operations management, sales planning <-> sales, team management <-> operations management] |

Possible Synonyms Problem

| | Actual | Predicted | Error Type | Detailed Error Type |
|---|---|---|---|---|
| 0 | | brand marketing | False Positive | False Positive |
| 1 | | social media | False Positive | False Positive |
| 2 | | web development | False Positive | False Positive |
| 3 | process development | | False Negative | False Negative |
| 4 | inspection | | False Negative | False Negative |
| 5 | | wireless networking | False Positive | False Positive |
| 6 | | logistics | False Positive | False Positive |
| 7 | | budgeting | False Positive | False Positive |
| 8 | testing | | False Negative | False Negative |
| 9 | defect identification | | False Negative | False Negative |
| 10 | | lesson planning | False Positive | False Positive |
| 11 | | teaching | False Positive | False Positive |
| 12 | | ui design | False Positive | False Positive |
| 13 | | ui design | False Positive | False Positive |
| 14 | interaction design | | False Negative | False Negative |

| | Skill | False Positive Count |
|---|---|---|
| 0 | operations management | 9 |
| 1 | contract law | 7 |
| 2 | sales | 7 |
| 3 | communication | 7 |
| 4 | logistics | 6 |
| 5 | legal compliance | 6 |
| 6 | ui design | 5 |
| 7 | research | 5 |
| 8 | brand marketing | 4 |
| 9 | computer networking | 4 |
| 10 | healthcare | 3 |
| 11 | taxation | 3 |
| 12 | social media | 3 |

Most Common False Positive Skills

That tells you your taxonomy contains **generic terms that are too broad**.

| | Skill | False Negative Count |
|---|---|---|
| 0 | network security | 4 |
| 1 | network monitoring | 4 |
| 2 | incident response | 4 |
| 3 | contract negotiation | 4 |
| 4 | defect identification | 3 |
| 5 | server-side logic | 3 |
| 6 | quality standards | 3 |
| 7 | customer feedback analysis | 3 |
| 8 | data analysis | 3 |
| 9 | auditing | 3 |
| 10 | testing | 3 |

This tells you which skills your taxonomy is **missing**.

## Outcome

Error analysis revealed the main weaknesses of the current taxonomy-based extraction system, particularly **false positives, missed skills, generic terms, and incomplete synonym mapping**. These findings can be used to improve the taxonomy and matching rules in the next stage.